2.3 Navigation Systems

Navigation with Indian Constellation (NavIC) is India's independent regional navigation satellite system catering to a coverage area of India and 1500 km beyond the Indian mainland. ISRO has established the space and ground infrastructure. ISRO is making continuous efforts to enable civilian sectors like land transportation, aviation, maritime, mapping, surveying, geodesy, timing, telecommunications, etc., to utilize the services offered by NavIC. GPS Aided Geo Augmented Navigation (GAGAN) is a space-based augmentation system for civil aviation purposes in the Indian Flight Information Region (FIR). ISRO has established the space segment, while the ground segment is established by the Airports Authority of India (AAI).

Major developments in navigation systems during 2025 have been:

2.3.1 NavIC base layer constellation

NavIC base layer currently consists of four satellites (IRNSS-1B, 1F, 1I, and NVS-01) providing PNT service. NVS-02 was successfully launched on 29th January 2025 onboard LVM3-M5. However, the satellite could not achieve its intended final orbit. It is capable of providing limited PNT service. Performance of indigenous atomic clock in NVS-02 has been thoroughly evaluated and it is exceeding the specification.

2.3.2 NavIC ground segment

NavIC ground segment is fully functional. Activities have progressed for establishment of reference stations (known as IRIMS) in foreign locations to improve the measurement baseline. Interference and noise surveys have been completed at South Africa and France and further activities have been initiated.

2.3.3 Navigation message authentication

Navigation message authentication technique has been developed and tests have been carried out. Software libraries have been developed for operationalization in NavIC SPS. Interface control definitions have also been worked out.

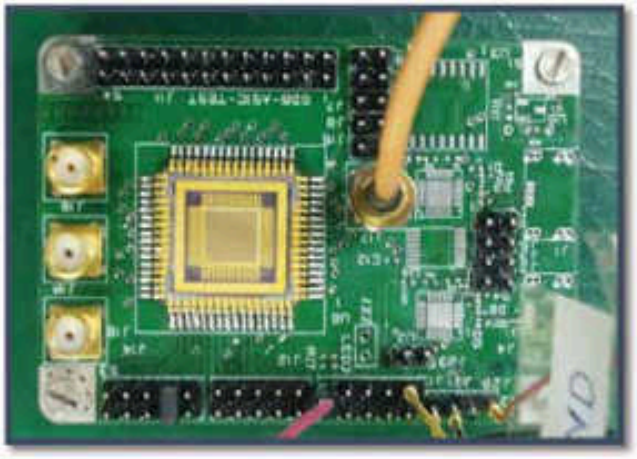
2.3.4 GNSS digital baseband (GDB) ASIC

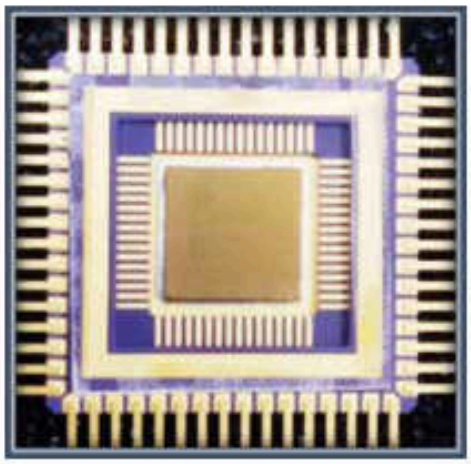
Indigenously designed all-in-view NavIC + GNSS digital baseband (GDB) ASIC was realised with 28 nm foundry node. The ASIC supports 100 tracking channels including NavIC & all GNSS open service signals. It has dual core processor and on-chip PLL for configurable processor frequency. It also incorporates multiple features like anti-jamming,

2.3

Navigation Systems

anti-spoofing, message authentication, pseudolite signal acquisition, and can be used for low power applications.





GDB ASIC

2.3.5 Efforts towards precise product generation

Post-processed NavIC products, needed for high accuracy applications involving NavIC is identified as a priority development. Under this effort, post-processed clock and orbit products were successfully used in remote sensing application on trial basis.

2.3.6 Time dissemination

Department of Legal Metrology, along with NPL and ISRO, is executing the project of dissemination of Indian Standard Time (IST) through five regional timescales and one disaster recovery centre. ISRO has carried out integration of the five regional timescales. Four timescales located in Ahmedabad, Bengaluru, Bhubaneshwar, and Faridabad have successfully completed site acceptance testing.



Time dissemination setup

2.4 Space Science Exploration and Research

The Department of Space's (DOS) Space Science and Exploration program achieved significant programmatic and scientific milestones, firming India's growing influence in several discovery-class of scientific results, and robustly progressing future exploratory ambitions. Operational missions delivered critical, high-quality data: the Aditya-L1 solar observatory expanded its public data archive, releasing over 20 TB for global utilization. The X-ray Polarimeter Satellite, XPoSat, marked key achievements, successfully completing its first genuine Target of Opportunity (ToO) observation, which resulted in the detection of a thermonuclear burst followed by a superburst from a neutron star source. XPoSat ceremonially released its maiden data set (143 GB) and announced the Guest Observer opportunity for the national community in October 2025. Concurrently, the operational AstroSat mission continued its observations, and brought out significant results. Chandrayaan-2 Orbiter continues to study the Moon and advanced data products such as global elemental maps and polar mosaics are made available in ISSDC, which will be useful for global lunar exploration. Strategic groundwork for future missions also accelerated: the collaborative Chandrayaan-5/LUPEX mission formally advanced with the signing of its Phase-B Implementation Arrangement (IA) with JAXA, while the Venus Orbiter Mission (VOM) secured international cooperation in Venusian science. VOM utilized this momentum by organizing a dedicated national science meet in October 2025 and releasing an Announcement of Opportunity (AO) to strengthen community involvement. Preparations for the Chandrayaan-4 lunar sample return mission intensified, marked by the "National Meet on Chandrayaan-4 Science" in April 2025, a critical step toward defining requirements for domestic sample analysis facilities. Furthermore, DOS advanced long-term strategic planning by synthesizing outcomes from the DOS Chintan Shivir 2025 and initiating the concept formulation for specialized future missions, including India's first dedicated magnetospheric mission.

This chapter will briefly present the major achievements of DOS in the sector of Space Science & Exploration.

2.4.1 Updates on the Active Science Missions

a. Chandrayaan-3

Launched on July 14, 2023, the Chandrayaan-3 mission successfully demonstrated soft-landing and roving on the Moon's Southern high latitude by 23 August 2023. After separating from the lander, the Propulsion Module (PM) was kept in lunar orbit until October 2023 before being maneuvered into a high-altitude Earth-bound orbit via Trans-Earth Injection (TEI). This interplay of gravity fields has led the spacecraft to enter the Moon's Sphere of Influence (SOI) on November 04, 2025. On November 06, 2025 07:23

UT, the first lunar flyby event took place outside the Indian Deep Space Network (IDSN) visibility at a distance of 3740 km from the Moon's surface. The second flyby event was visible from the IDSN, the closest approach distance was 4537 km from the Moon's surface on November 11, 2025, 23:18 UT. CH3-PM exited the Moon's SOI on November 14, 2025.

Meanwhile, there have been several significant scientific results from the post-analysis of the Chandrayaan-3 datasets.

b. AstroSat

AstroSat, India's multi-wavelength space-based astronomical observatory, remains fully operational with all scientific payloads working satisfactorily. AstroSat data has led to over 532 refereed publications globally. The mission provides highest-resolution UV imaging and detected X-ray polarization from celestial objects. Data accessibility improved after the successful completion of testing and evaluation for porting the extensive AstroSat archives to the PRADAN portal. The mission continues to support research, contributing to 47 PhD theses and funding data utilization projects.

c. Chandrayaan-2

The Chandrayaan-2 orbiter continues its sustained scientific exploration from a 100 km lunar polar circular orbit. The mission continues to yield foundational data on lunar volatiles and composition, with 48 TB of data released to over 10,000 users. Observations on the effects of Coronal Mass Ejections from the Sun on the lunar exosphere is a significant result and first of its kind. Efforts are underway to publish a special issue on the orbiter's six-year achievements in a journal.

d. Aditya-L1

Aditya-L1, India's dedicated solar observatory, has released over 20 TB of scientific data for global utilization as of October 2025. The mission successfully serviced several Target of Opportunity (ToO) and calibration proposals. Community engagement remains robust: the 11th hands-on data training workshop was successfully held in October 2025. A joint ISRO-ESA Heliophysics Workshop is scheduled for January 2026, which will promote collaborative data analysis.

e. XPoSat

XPoSat reached a major milestone with the ceremonial release of 143 GB of maiden datasets to the scientific community in October 2025. The XSPECT payload achieved a key scientific outcome by detecting a thermonuclear Type-I X-ray burst immediately followed by a rare, long-duration superburst from the neutron star binary system 4U 1608-52.

2.4

Space Science Exploration and Research

This was confirmed as the first genuine Target of Opportunity observation for XSPECT. The mission continues regular observation campaigns of astronomical sources like Sco-X1. Furthermore, the XPoSat Proposal Processing System (XPPS) was deployed on an internet platform, enabling national guest observers to submit XSPECT-based science proposals, following the National Science Meet held for data release and performance appraisal.

2.4.2 Updates on Newly Approved Space Science Missions

a. Chandrayaan-4

Chandrayaan-4, India's forthcoming Lunar Sample Return Mission, is designed to collect and return lunar samples from the polar region of the Moon to Earth for detailed laboratory analysis. The science team has reviewed potential landing sites, prioritizing them based on scientific potential using existing lunar datasets, including Chandrayaan-2. Preparations are in progress for firming up the plan for sample analysis and archival facilities.

b. Chandrayaan-5 / LuPEX

The Chandrayaan-5 / LUPEX mission, a joint venture between ISRO and JAXA targeting the in-situ study of lunar polar volatiles, secured Financial Sanction (FS) on March 10, 2025. The mission took a significant step forward when the Phase-B Implementation Arrangement (IA) detailing agency responsibilities was formally signed on August 29, 2025. Technical Interface Meetings (TIMs) were held in May and August 2025. The Science Working Group (SWG) has been constituted to maximize the science returns from the lander and rover payloads, and the Principal Scientist of the mission has also been identified. The Preliminary Design Review (PDR) of the mission has also been completed.

c. Venus Orbiter Mission

The objectives of the Venus Orbiter Mission (VOM) encompass studying the Venusian atmosphere, ionosphere, surface, subsurface, and its interaction with the Sun. International collaboration efforts have been formalized, with approval granted for MoUs related to key international payloads. The Science Working Group (SWG) has been constituted to maximize the science returns from the payloads, and the Principal Scientist of the mission has also been declared. The Preliminary Design Review (PDR) of the mission has also been completed.

2.4.3 Space Science Missions in Approval and Study Phases

The Mars Lander Mission (MLM), as the follow-on of India's Mars Orbiter Mission (MOM), has obtained the approval of the Space Commission, and is currently under approval by

the Cabinet. The mission is aimed as scientific studies of the Martian surface, near-surface environment, as well as Sun-Mars interactions.

The DISHA (Disturbed and Quiet Time Ionosphere-Thermosphere System at High Altitudes) mission is a proposed twin-satellite system designed to study the effects of solar forcing on the Earth's upper atmosphere. Its core objective is to characterize the ionosphere-thermosphere system by simultaneously measuring neutral and charged particle parameters. This includes understanding variability caused by lower atmospheric processes and solar storm-induced geomagnetic disturbances. DISHA-H (High inclination, > 85°) and DISHA-L (Low inclination, ~25°), both at ~ 400 km altitude, aim to decouple latitudinal and longitudinal variations for effective space weather forecasting. DISHA is currently in the process of obtaining internal clearances.

The ExoWorlds mission is a proposed space-based observatory, based on the transiting spectrometry technique from the Sun-Earth L2 point, intended for the study of the atmosphere of exoplanets. Its central goal is to conduct a comprehensive survey of the chemical composition and atmospheric processes of a large population of exoplanets, leading to a new understanding and classification of planetary systems. ExoWorlds is also currently in the process of obtaining internal clearances.

The Department of Space is also formulating concepts for astronomy missions like DAKSHA, PRATUSH, SEAMS, and INSIST. Furthermore, discussions are underway with the scientific community for India's participation in space-based deci-Hz Gravitational Wave (GW) missions. Continuous engagement is maintained with the scientific community for generating concept notes on missions targeting the Cosmic Dark Ages and Cosmic Dawn.

2.4.4 Major Scientific Results

a. First in-situ temperature measurements near southern polar region of the Moon by ChaSTE experiment onboard Chandrayaan-3 Vikram lander

Measurements from Chandra's Surface Thermophysical Experiment (ChaSTE) payload on Vikram Lander onboard Chandrayaan-3 the first-ever in-situ measurements of the temperature profile was made and thermophysical properties within the top 10 cm of the lunar surface at a high latitude south polar landing location. Publication: Nature: Communications Earth & Environment (https://doi.org/10.1038/s43247-025-02114-6).

(https://www.isro.gov.in/India_Bags_Credit_First-Ever_In-Situ_Measurement.html)

2.4

Space Science Exploration and Research

b. Primitive mantle material revealed by Chandrayaan-3 APXS volatile measurements in the South Polar Region of the Moon

Concentrations of volatile elements measured at Shiv Shakti station near the South Polar Region were determined using PRL's Alpha Particle X-ray Spectrometer (APXS) onboard the Pragyan rover of the Chandrayaan-3 mission. This new finding makes the Chandrayaan-3 landing site a promising site to access primitive mantle samples, which is otherwise lacking in the existing lunar collections.

(https://www.isro.gov.in/ISRO_EN/Ch3_APXS_volatile_measurements.html)

Publication: Nature: Communications Earth & Environment
(https://doi.org/10.1038/s43247-025-02305-1)
Publication: Nature: Scientific Reports (https://doi.org/10.1038/s41598-025-21988-2)

c. Observation of substantially high and variable electron density in the near-Surface environment of the Moon, by the RAMBHA-LP experiment onboard Chandrayaan-3 lander

The Chandrayaan-3 mission used its RAMBHA-LP instrument to take measurements of the near-surface plasma environment at the Shiv Shakti point; the first-ever plasma measurements at lunar Southern high latitudes. It is observed that the electron density near the lunar surface varied between 380 to 600 electrons per cubic centimeter, the variations being controlled by two main factors: the solar wind (when the Moon is facing the Sun) and charged particles escaping from Earth's magnetic field (when the Moon is behind Earth, within the geomagnetic tail). The electrons were found to be energetic, with kinetic temperatures in the range of 3000-8000 K, indicating non-thermal processes. The study also highlighted the role of molecular ions, in addition to the elemental ions, in creating this plasma near the Moon.

Publication: Monthly Notices of the Royal Astronomical Society (https://doi.org/10.1093/mnras/staf1276)

d. Observation of the effects of solar Coronal Mass Ejection (CME) on the Moon's sunlit exosphere, by CHACE-2 onboard Chandrayaan-2

The Chandrayaan-2 lunar orbiter has made the first-ever observation of the effects of the Sun's Coronal Mass Ejection (CME) on the Moon, with one of its scientific instruments Chandra's Atmospheric Composition Explorer-2 (CHACE-2) onboard. Observations from

2.5 Space Transportation Systems

India's journey in launch vehicle development began in the 1960s with sounding rockets for atmospheric and meteorological studies. The first Indian-made sounding rocket, RH-75 (Rohini-75) took off in 1967 from the Thumba Equatorial Rocket Launching Station (TERLS) in Thiruvananthapuram which laid the groundwork for mastering rocket technology. India has achieved Atmanirbharta in space transportation systems through operationalization of the satellite launch vehicles viz. Polar Satellite Launch Vehicle (PSLV), Geosynchronous Satellite Launch Vehicle (GSLV), and Geosynchronous Satellite Launch Vehicle – Mark III (LVM3). The development of Small Satellite Launch Vehicle (SSLV), capable of launching small satellites of mass up to 500kg to Low-Earth Orbits has been completed. The new Space Policy introduced by the Government of India in 2023 wherein the operational space transportation systems will be realized by Indian industry through NSIL. In view of this, the realization of PSLV has already been commenced through industry and technology transfer of SSLV to Industry has been initiated by the Department. The POEM platform in PSLV has opened avenues for start-ups to demonstrate their various payloads under the ambit of the New Space Policy.

Self-reliance in space transportation system has been a key element in framing India's future space missions. This portfolio of launch vehicles has enabled India to undertake multi-orbit and multi-satellite missions for accessing near-earth orbits. Apart from launching satellites which find societal benefits, India is now having the expanded vision with Human Space Programme and further space explorations. A human rated space transportation system based on LVM3 configuration (HLVM3) has been qualified and is ready for undertaking the first un-manned Gaganyaan mission. Development of the Next Generation Launch Vehicle (NGLV) with reusable technology has been initiated to meet the expanded vision of Indian Space programme. Government of India approved the Establishment of the Third Launch Pad at Sriharikota for the Next Generation launch vehicles of ISRO, support as standby launch pad for the SLP, and also enhance the launch capacity for future Indian human spaceflight missions. The development of advanced propulsion technologies including LOX-Methane propulsion, Semi-cryogenic propulsion, Electric propulsion and Air-breathing propulsion have been taken up. Technologies for Vertical Take-off & Vertical Landing and Winged body orbital re-entry vehicles, are also being developed by ISRO with the support of industry and academia.

Major Events

2.5.1 Polar Satellite Launch Vehicle (PSLV)

Polar Satellite Launch Vehicle (PSLV) completed its 63rd launch during the period and continued to demonstrate its reliability and versatility through multi-satellite and multi-orbit missions.

a. PSLV-C61 vehicle lifted off with EOS-09 spacecraft on May 18, 2025 at 05:59Hrs from SDSC, Sriharikota. The performance of PSLV-C61 vehicle was normal till the 2nd stage and an anomaly was noticed during the functioning of 3rd stage solid motor & the mission could not be accomplished. A national level committee was constituted to investigate the anomaly in the performance observed & identify the root cause of the failure and suggest corrective measures to be implemented. The committee had exhaustive deliberations and has submitted the findings & recommendations. Based on the recommendations of the National Level Committee comprising of eminent experts from academia & industry, the third stage of PSLV i.e., HPS3 motor with modified design was realised and two static tests were successfully completed on October 06, 2025 and November 19, 2025 as in flight, from SDSC, SHAR. The overall performance of the motor and subsystems were as expected and closer to nominal performance.



First Static test of the modified HPS3 Motor conducted on October 6, 2025



Second Static test of the modified HPS3 Motor conducted on November 19, 2025

b. PSLV-C62/ EOS-N1 mission will launch the EOS-N1 spacecraft which is being realised by ISRO. The launch vehicle stacking activities for the mission has commenced and the mission is tentatively planned during fourth quarter of 2025-26.

c. PSLV-C63/ TDS-01 Mission will launch the Technology Demonstration Satellite (TDS-01) of ISRO for the demonstration of various technologies including Quantum Technology Demonstrator payload, Traveling Wave Tube Amplifier and high thrust Electrical Propulsion Systems. The mission is tentatively planned during fourth quarter of 2025-26.

2.5

Space Transportation Systems

d. EOS-10 satellite of the Oceansat series & Indo-Mauritius Joint Satellite (IMJS-1) will be launched by the first PSLV realised by Indian Industry, undertaking end-to-end responsibility of stage realisation. The launch is tentatively scheduled during fourth quarter of 2025-26.

2.5.2 Geosynchronous Satellite Launch Vehicle (GSLV)

GSLV is a three-stage vehicle with solid, liquid, and cryogenic upper stage, designed to place a 2000 kg class of spacecraft into Geosynchronous Transfer Orbit (GTO).

a. GSLV-F16/ NISAR mission launched the NASA-ISRO Synthetic Aperture Radar Satellite, which is an advanced Earth Observation satellite jointly developed by NASA & ISRO. The mission was successfully accomplished on July 30, 2025 at 17:40 Hrs from the Second Launch Pad (SLP) at SDSC, Sriharikota. This was the first LEO mission of GSLV launching NISAR spacecraft to a Sun Synchronous Polar Orbit. NISAR is the first of its kind mission, jointly developed by ISRO and NASA for microwave imaging purpose globally. It carries an L-band and S-band Synthetic Aperture Radar (SAR), with capability to acquire fully polarimetric and interferometric data.



GSLV-F16/ NISAR Mission

b. GSLV F17/ EOS-05 Mission will launch the EOS-05 spacecraft which is a GEO-imaging satellite. Realisation of vehicle stages systems are being carried and the satellite is planned to be launched from SDSC SHAR during fourth quarter of 2025-26.

c. GSLV F18/ NVS-03 Mission will launch the third NVS satellite (NVS-03) in a series of 5 navigation satellites intended to replace/ augment the aging in-orbit NavIC constellation and is tentatively scheduled during fourth quarter of 2025-26.

2.5.3 Geosynchronous Satellite Launch Vehicle MKIII (LVM3)

LVM3 is a three-stage launch vehicle with two solid strap-on motors (S200), one liquid core stage (L110) and a cryogenic upper stage (C25).

a. LVM3-M5/ CMS-03 mission successfully launched the CMS-03 spacecraft on

2.6 Gaganyaan

Gaganyaan is a national programme to demonstrate the capability to launch human beings to low earth orbit on an Indian launch vehicle and bring them back safely to earth. The mission is broadly divided into three major phases, namely, ascent phase, orbital phase and descent phase.

In the ascent phase, the human rated launch vehicle carries the Orbital Module to a low earth orbit. The orbital phase starts with the injection of the Orbital Module by the launch vehicle into an elliptical orbit. This is further raised to a circular orbit using the engines present in the Orbital Module. The orbital phase ends with the beginning of the de-boost maneuvers carried out to initiate the return journey. The descent phase begins with the de-boost maneuvers which sets the course of the module towards the designated touch-down location. A series of activities will be carried out during the descent phase, which will finally end with a low-velocity splash down at a designated location in sea waters.

The Gaganyaan programme involves complex and multi-disciplinary activities with an emphasis on human-centric approach in designing, realizing and testing various subsystems.

Various precursor missions are planned for demonstrating the Technology Preparedness Levels before carrying out the actual Human Space Flight mission. These demonstrator missions include Integrated Air Drop Test (IADT), Pad Abort Test (PAT) and Test Vehicle (TV) flights. As a precursor to crewed mission H1, three uncrewed missions G1, G2 and G3 are planned towards end-to-end testing of all systems involved. The first uncrewed mission G1 is configured as an experimental mission to enable early flight testing of various systems.

2.6.1 Gaganyaan First uncrewed Mission

a. Human Rated Launch Vehicle (HLVM3)

HLVM3 vehicle consists of three propulsion modules (HS200 strap-ons, L110 stage, C32 stage) and Crew Escape System (CES). Vehicle integration activities commenced last year. Further, stacking of both HS200 solid motors completed. L110 and C32 stages are ready and under preservation. Avionics flight packages for the Solid Strap-on Nose Cone realized and delivered for integration. Telemetry and Telecommand packages qualified and delivered. Navigation and Guidance Control and power checks completed. Instrumentation checks of inter-stages and equipment bay completed.

2.14 India's Space Vision 2047

Government of India has announced the Space Vision 2047, which targets establishing Bharatiya Antariksh Station (BAS) by 2035 and landing of an Indian on Moon by 2040. Towards this, Government has approved five important projects: Gaganyaan follow-on missions and establishment of BAS 1st module by 2028, Development of Next Generation Satellite Launch Vehicle (NGLV) by 2032, Chandrayaan-4 by 2027, to develop and demonstrate the technologies to come back to Earth after successfully landing on the Moon and also collect moon samples, Venus Orbiter Mission (VOM) by 2028, to study the Venusian surface and subsurface, atmospheric processes and influence of Sun on Venusian Atmosphere and establishment of Third Launch Pad (TLP) at Spaceport of India, Sriharikota.

ISRO has formulated a roadmap for space science exploration missions, integrating multiple domains of developments, towards achieving the goal of Space Vision 2047. These efforts focus on technological advancements, international partnerships, increasing the participation of private players and advancing space exploration missions.

The major milestones of the achieving Space Vision 2047 are Launch of 1st module of Bharatiya Antariksh Station (BAS) by 2028, Establishment of full BAS by 2035 and Indian Moon landing by 2040.



5.1

Milestones

second- generation satellites envisaged for the Navigation with Indian Constellation (NavIC) services. NVS series of satellites will sustain and augment the NavIC with enhanced features and incorporate L1 band signals additionally to widen the services.

- On July 14, 2023, the LVM-3 M4 vehicle, in its fourth operational flight, launched Chandrayaan-3 to a precise Geo Transfer Orbit (GTO).
- On July 30, 2023, PSLV-C56 launched DS-SAR satellite, along with 6 co-passengers. Configured in its core-alone mode, C56 launched a 360 kg satellite into a Near-Equatorial Orbit (NEO) at 5 degrees' inclination and 535 km altitude.
- On August 5, 2023, following a series of manoeuvres, Chandrayaan-3 was inserted into the lunar orbit. The lander module was successfully separated from the propulsion module on August 17, 2023. The lander module was brought to a 25 km x 134 km orbit around the moon on August 20, 2023.
- On August 23, 2023, Chandrayaan-3 soft-landed on the moon.
- On August 24, 2023, the rover descended on the lunar surface. For over 14 Earth days, in-situ scientific experiments were conducted on the moon, near its south pole.
- On September 03, 2023, a hop test was conducted on Chandrayaan-3 lander. On October 13, 2023, the propulsion module was brought to an Earth-bound orbit, demonstrating the capability of Earth-return manoeuvres.
- On September 2, 2023, PSLV-C57 launched Aditya L1, the first Indian solar observatory, into precise orbit. Four manoeuvres and a planned trajectory correction manoeuvre ensured Aditya-L1's Trans-Lagrangean Point 1 Insertion (TL1I) on October 8, 2023.
- On October 18, 2023, the Gaganyaan TV D1 Test Flight was accomplished. The Crew Escape System performed as intended. Performance of various separation systems incorporated into the Crew Escape System, the characteristics and deceleration systems demonstration at higher altitude & its recovery was demonstrated

2024

- The Aditya-L1 observatory was successfully injected into Halo orbit L1 on January 6, 2024
- PSLV-C58 successfully launched XPoSat, (X-ray Polarimeter Satellite) India's first dedicated polarimetry mission to study various dynamics of bright astronomical X-ray sources in extreme conditions on January 1, 2024, into the intended orbit. Both the XSPECT and POLIX payloads began astronomical observations during January and February respectively
- PSLV-C58 had the PSLV Orbital Experimental Module-3 (POEM-3) that successfully conducted 10 experiments through payloads identified and supplied by ISRO and IN-SPACe

- On February 17, 2024, GSLV-F14 successfully deployed INSAT-3DS, a meteorological satellite into the Geosynchronous Transfer Orbit (GTO). GSLV-F14/INSAT-3DS mission is fully funded by the Ministry of Earth Sciences (MoES). The satellite initiated Earth imaging operations with the first set of images captured on March 7, 2024
- On March 22, 2024, the Reusable Launch Vehicle Autonomous Landing Mission (RLV LEX-02) landing experiment, the second of the series, was conducted at Aeronautical Test Range (ATR), Chitradurga in Karnataka. After accomplishment of RLV-LEX-01 mission in last year, RLV-LEX-02 demonstrated the autonomous landing capability of RLV from off-nominal initial conditions at release from Helicopter. The RLV was made to undertake more difficult manoeuvres with dispersions, correct both cross-range and downrange and land on the runway in a fully autonomous mode.
- On June 23, 2024, Reusable Launch Vehicle Autonomous Landing Mission (RLV LEX - 03) was conducted at the Aeronautical Test Range (ATR) in Chitradurga, Karnataka. RLV LEX-03 re-demonstrated the autonomous landing capability of the RLV under more challenging release conditions (cross range of 500 m against 150 m for LEX-02) and more severe wind conditions.
- On July 22, 2024, the second experimental flight for the demonstration of Air Breathing Propulsion Technology was carried out. The Propulsion systems were symmetrically mounted on either side of a RH-560 Sounding rocket and launched from SDSC, Sriharikota. The flight test achieved satisfactory performance of the Sounding Rocket along with successful ignition of the Air Breathing propulsion systems.
- On August 16, 2024, the third developmental flight of SSLV was successfully launched. The SSLV-D3, in its third and final development flight, launched EOS-08 satellite.
- On Nov 19, 2024, GSAT-N2 satellite was successfully launched on-board Falcon-9 of M/s SpaceX, USA.
- On Dec 05, 2024, PSLV-C59 vehicle launched Proba-3 satellite. This is a dedicated commercial mission conducted through NSIL, featuring the launch of the Proba-3 satellite for an In-Orbit Demonstration (IOD) mission aimed at showcasing precise formation flying. The mission comprised of two spacecrafts viz., the Coronagraph Spacecraft (CSC) and the Occulter Spacecraft (OSC), both launched together in a stacked configuration.
- On December 30, 2024, PSLV-C60 vehicle successfully launched SpaDeX satellites. SpaDeX mission is a cost-effective technology demonstrator mission for the demonstration of in-space docking. The SpaDeX mission consists of two small spacecraft (about 220 kg each) which are launched independently and simultaneously, into a 470 km circular orbit at 55° inclination.
- PSLV-C60 had the PSLV Orbital Experimental Module-4 (POEM-4) that carried a total of 24 payloads, of which 14 payloads are from ISRO/DOS Centres and 10 payloads are

5.1

Milestones

from various Non-Government Entities (NGEs) comprising Academia and Start-ups that have been received through IN-SPACe.

2025

- On January 6, 2025, ISRO shared the first batch of scientific data from Aditya-L1 -India's pioneering solar observatory - with scientists worldwide at ISRO Headquarters in Bengaluru.
- On January 16, 2025, ISRO successfully completed docking of two SPADEX satellites (SDX-01 & SDX-02). After the first docking, undocking, and circumnavigation, the two spacecrafts were docked again on April 20, 2025. The spacecraft then switched to combined control mode, and power transfer between them was tested and completed successfully on April 21, 2025.
- On January 21, 2025, a Public-Private Partnership (PPP) was initiated by formal agreement for establishing 12 Earth Observation satellite constellation in order to enhance India's satellite surveillance capabilities.
- On January 29, 2025, GSLV-F15 vehicle launched NVS-02 satellite weighing about 2250 kg, into a Geosynchronous Transfer Orbit. This launch marked the 100th Launch from the India's Spaceport - Sriharikota.
- ISRO developed the first "Make-In-India" 32-bit Microprocessors for space applications jointly with SCL, Chandigarh. On March 5, 2025, the first batches of two 32-bit microprocessors for space use were handed to Department of Space (DoS) by Ministry of Electronics and Information Technology (MeitY).
- On March 27, 2025, ISRO successfully completed the life test of 1000hrs on the 300mN Stationary Plasma Thruster, that is developed for induction into the Electric Propulsion System of satellites.
- On May 18, 2025, PSLV-C61 mission, ISRO's 101st launch endeavour, was attempted and performance was normal till 2nd stage. Due to an observation in 3rd stage, the mission could not be accomplished.
- During March to May, 2025, a series of hot tests for Semi-cryo engine Power Head Test Article (PHTA) were successfully completed.
- On June 16, 2025, Honourable Minister of Home Affairs of India, Shri Amit Shah inaugurated the Integrated Control Room for Emergency Response (ICR-ER). It is established as a national command center using ISRO's satellite data for real time disaster management.
- On June 25, 2025, Axiom-04 mission was launched by SpaceX with ISRO astronaut Shubhanshu Shukla aboard a Dragon spacecraft. During the historic 18-day scientific stay at the International Space Station (ISS), Group Captain Shubhanshu Shukla completed a suite of seven microgravity experiments aimed at enhancing understanding of human spaceflight and microgravity science and returned back to earth safely.

- On July 30, 2025, NASA-ISRO Synthetic Aperture Radar (NISAR), the first joint satellite of ISRO and NASA was launched successfully by ISRO's GSLV-F16 Mission from Satish Dhawan Space Centre, Sriharikota.
- On August 8, 2025, first static test of the KALAM 1200 Motor - the first stage of Vikram-1 Launch Vehicle of a private startup, M/s Skyroot Aerospace Pvt. Ltd. (SAPL) is accomplished successfully by ISRO in Satish Dhawan Space Centre, Sriharikota.
- On August 24, 2025, ISRO accomplished first Integrated Air Drop Test (IADT-01) for Gaganyaan program at Satish Dhawan Space Centre (SDSC), Sriharikota.
- On August 27, 2025, Dr. V. Narayanan, Chairman, ISRO / Secretary, Department of Space laid the foundation for the Launch Pad at the SSLV Launch Complex (SLC) at Kulasekarapattinam, Tuticorin Dt., Tamil Nadu.
- On September 10, 2025, a technology transfer agreement was signed between NewSpace India Limited, ISRO, IN-SPACe & Hindustan Aeronautics Limited (HAL) for the transfer of the Small Satellite Launch Vehicle (SSLV) technology at ISRO Headquarters, Bangalore.
- On November 02, 2025, LVM3 launch vehicle successfully launched the heaviest Indian communication satellite, CMS-03, a multi-band communication satellite, into Geosynchronous Transfer Orbit (GTO). This mission (LVM3-M5) also successfully achieved the first-ever in-space restart of its indigenously developed C25 cryogenic upper stage, powered by the CE-20 engine.
- Physical Research Laboratory (PRL), Ahmedabad, discovered a new exoplanet, TOI-6038A b, having a dense sub-Saturn size with a mass of 78.5 Earth masses and a radius of 6.41 Earth radii in a wide binary system. The planet orbits a bright, metal-rich F-type star every 5.83 days in a circular orbit.
- On December 24, 2025, LVM3 launch vehicle successfully launched the BlueBird Block-2 communication satellite of AST SpaceMobile, USA into its intended orbit. This mission marks the 6th operational flight of LVM3.

QUESTIONS

1. Why do HCl, HNO_3, etc., show acidic characters in aqueous solutions while solutions of compounds like alcohol and glucose do not show acidic character?
2. Why does an aqueous solution of an acid conduct electricity?
3. Why does dry HCl gas not change the colour of the dry litmus paper?
4. While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid?
5. How is the concentration of hydronium ions (H_3O^+) affected when a solution of an acid is diluted?
6. How is the concentration of hydroxide ions (OH^-) affected when excess base is dissolved in a solution of sodium hydroxide?

2.3 HOW STRONG ARE ACID OR BASE SOLUTIONS?

We know how acid-base indicators can be used to distinguish between an acid and a base. We have also learnt in the previous section about dilution and decrease in concentration of H^+ or OH^- ions in solutions. Can we quantitatively find the amount of these ions present in a solution? Can we judge how strong a given acid or base is?

We can do this by making use of a universal indicator, which is a mixture of several indicators. The universal indicator shows different colours at different concentrations of hydrogen ions in a solution.

A scale for measuring hydrogen ion concentration in a solution, called pH scale has been developed. The p in pH stands for '*potenz*' in German, meaning power. On the pH scale we can measure pH generally from 0 (very acidic) to 14 (very alkaline). pH should be thought of simply as a number which indicates the acidic or basic nature of a solution. Higher the hydronium ion concentration, lower is the pH value.

The pH of a neutral solution is 7. Values less than 7 on the pH scale represent an acidic solution. As the pH value increases from 7 to 14, it represents an increase in OH^- ion concentration in the solution, that is, increase in the strength of alkali (Fig. 2.6). Generally paper impregnated with the universal indicator is used for measuring pH.

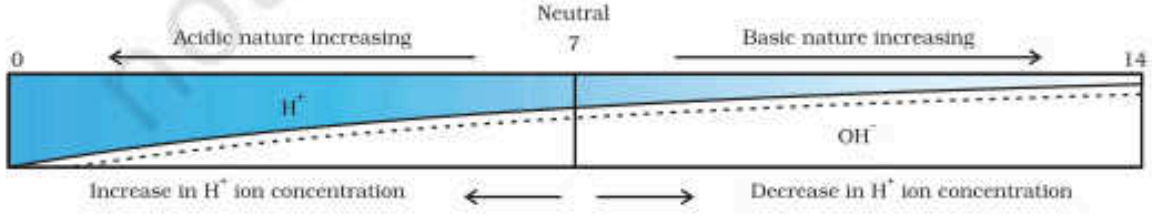


Figure 2.6 *Variation of* pH *with the change in concentration of* H^+(aq) *and* OH^-(aq) *ions*

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Activity 2.11

- Test the pH values of solutions given in Table 2.2.
- Record your observations.
- What is the nature of each substance on the basis of your observations?

Table 2.2

S. No.	Solution	Colour of pH paper	Approx-imate pH value	Nature of substance
1	Saliva (before meal)			
2	Saliva (after meal)			
3	Lemon juice			
4	Colourless aerated drink			
5	Carrot juice			
6	Coffee			
7	Tomato juice			
8	Tap water			
9	1M NaOH			
10	1M HCl			

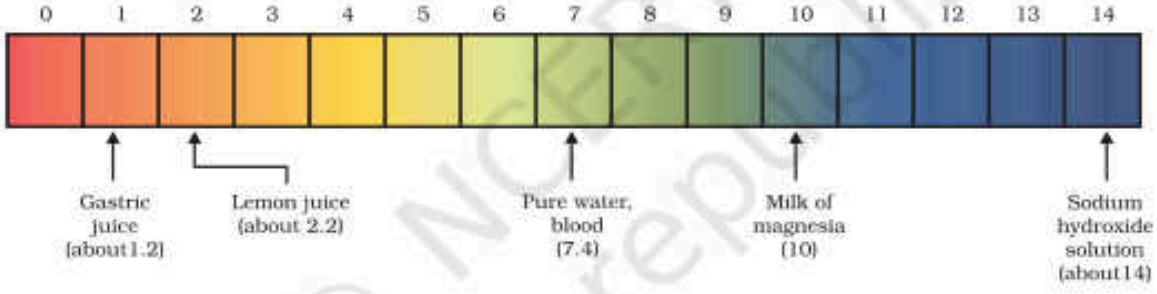


Figure 2.7 pH *of some common substances shown on a* pH *paper (colours are only a rough guide)*

The strength of acids and bases depends on the number of H^+ ions and OH^- ions produced, respectively. If we take hydrochloric acid and acetic acid of the same concentration, say one molar, then these produce different amounts of hydrogen ions. Acids that give rise to more H^+ ions are said to be strong acids, and acids that give less H^+ ions are said to be weak acids. Can you now say what weak and strong bases are?

2.3.1 Importance of pH in Everyday Life

Are plants and animals pH sensitive?

Our body works within the pH range of 7.0 to 7.8. Living organisms can survive only in a narrow range of pH change. When pH of rain water is less than 5.6, it is called acid rain. When acid rain flows into the rivers, it lowers the pH of the river water. The survival of aquatic life in such rivers becomes difficult.

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Do You Know?

Acids in other planets

The atmosphere of venus is made up of thick white and yellowish clouds of sulphuric acid. Do you think life can exist on this planet?

What is the pH of the soil in your backyard?

Plants require a specific pH range for their healthy growth. To find out the pH required for the healthy growth of a plant, you can collect the soil from various places and check the pH in the manner described below in Activity 2.12. Also, you can note down which plants are growing in the region from which you have collected the soil.

Activity 2.12

- Put about 2 g soil in a test tube and add 5 mL water to it.
- Shake the contents of the test tube.
- Filter the contents and collect the filtrate in a test tube.
- Check the pH of this filtrate with the help of universal indicator paper.
- What can you conclude about the ideal soil pH for the growth of plants in your region?

pH in our digestive system

It is very interesting to note that our stomach produces hydrochloric acid. It helps in the digestion of food without harming the stomach. During indigestion the stomach produces too much acid and this causes pain and irritation. To get rid of this pain, people use bases called antacids. One such remedy must have been suggested by you at the beginning of this Chapter. These antacids neutralise the excess acid. Magnesium hydroxide (Milk of magnesia), a mild base, is often used for this purpose.

pH change as the cause of tooth decay

Tooth decay starts when the pH of the mouth is lower than 5.5. Tooth enamel, made up of calcium hydroxyapatite (a crystalline form of calcium phosphate) is the hardest substance in the body. It does not dissolve in water, but is corroded when the pH in the mouth is below 5.5. Bacteria present in the mouth produce acids by degradation of sugar and food particles remaining in the mouth after eating. The best way to prevent this is to clean the mouth after eating food. Using toothpastes, which are generally basic, for cleaning the teeth can neutralise the excess acid and prevent tooth decay.

Self defence by animals and plants through chemical warfare

Have you ever been stung by a honey-bee? Bee-sting leaves an acid which causes pain and irritation. Use of a mild base like baking soda on the stung area gives relief. Stinging hair of nettle leaves inject methanoic acid causing burning pain.

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Do You Know?

Nature provides neutralisation options

Nettle is a herbaceous plant which grows in the wild. Its leaves have stinging hair, which cause painful stings when touched accidentally. This is due to the methanoic acid secreted by them. A traditional remedy is rubbing the area with the leaf of the dock plant, which often grows beside the nettle in the wild. Can you guess the nature of the dock plant? So next time you know what to look out for if you accidentally touch a nettle plant while trekking. Are you aware of any other effective traditional remedies for such stings?



Table 2.3 Some naturally occurring acids

Natural source	Acid	Natural source	Acid
Vinegar	Acetic acid	Sour milk (Curd)	Lactic acid
Orange	Citric acid	Lemon	Citric acid
Tamarind	Tartaric acid	Ant sting	Methanoic acid
Tomato	Oxalic acid	Nettle sting	Methanoic acid

QUESTIONS

1. You have two solutions, A and B. The pH of solution A is 6 and pH of solution B is 8. Which solution has more hydrogen ion concentration? Which of this is acidic and which one is basic?
2. What effect does the concentration of $H^+(aq)$ ions have on the nature of the solution?
3. Do basic solutions also have $H^+(aq)$ ions? If yes, then why are these basic?
4. Under what soil condition do you think a farmer would treat the soil of his fields with quick lime (calcium oxide) or slaked lime (calcium hydroxide) or chalk (calcium carbonate)?

2.4 MORE ABOUT SALTS

In the previous sections we have seen the formation of salts during various reactions. Let us understand more about their preparation, properties and uses.

2.4.1 Family of Salts

Activity 2.13

- Write the chemical formulae of the salts given below.
  Potassium sulphate, sodium sulphate, calcium sulphate, magnesium sulphate, copper sulphate, sodium chloride, sodium nitrate, sodium carbonate and ammonium chloride.

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- Identify the acids and bases from which the above salts may be obtained.
- Salts having the same positive or negative radicals are said to belong to a family. For example, NaCl and Na_2SO_4 belong to the family of sodium salts. Similarly, NaCl and KCl belong to the family of chloride salts. How many families can you identify among the salts given in this Activity?

2.4.2 pH of Salts

Activity 2.14

- Collect the following salt samples – sodium chloride, potassium nitrate, aluminium chloride, zinc sulphate, copper sulphate, sodium acetate, sodium carbonate and sodium hydrogencarbonate (some other salts available can also be taken).
- Check their solubility in water (use distilled water only).
- Check the action of these solutions on litmus and find the pH using a pH paper.
- Which of the salts are acidic, basic or neutral?
- Identify the acid or base used to form the salt.
- Report your observations in Table 2.4.

Salts of a strong acid and a strong base are neutral with pH value of 7. On the other hand, salts of a strong acid and weak base are acidic with pH value less than 7 and those of a strong base and weak acid are basic in nature, with pH value more than 7.

Table 2.4

Salt	pH	Acid used	Base used

2.4.3 Chemicals from Common Salt

By now you have learnt that the salt formed by the combination of hydrochloric acid and sodium hydroxide solution is called sodium chloride. This is the salt that you use in food. You must have observed in the above Activity that it is a neutral salt.



Seawater contains many salts dissolved in it. Sodium chloride is separated from these salts. Deposits of solid salt are also found in several parts of the world. These large crystals are often brown due to impurities. This is called rock salt. Beds of rock salt were formed when seas of bygone ages dried up. Rock salt is mined like coal.

You must have heard about Mahatma Gandhi's *Dandi March*. Did you know that sodium chloride was such an important symbol in our struggle for freedom?

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Common salt — A raw material for chemicals

The common salt thus obtained is an important raw material for various materials of daily use, such as sodium hydroxide, baking soda, washing soda, bleaching powder and many more. Let us see how one substance is used for making all these different substances.

Sodium hydroxide

When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. The process is called the chlor-alkali process because of the products formed– chlor for chlorine and alkali for sodium hydroxide.

$$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$$

Chlorine gas is given off at the anode, and hydrogen gas at the cathode. Sodium hydroxide solution is formed near the cathode. The three products produced in this process are all useful. Figure 2.8 shows the different uses of these products.

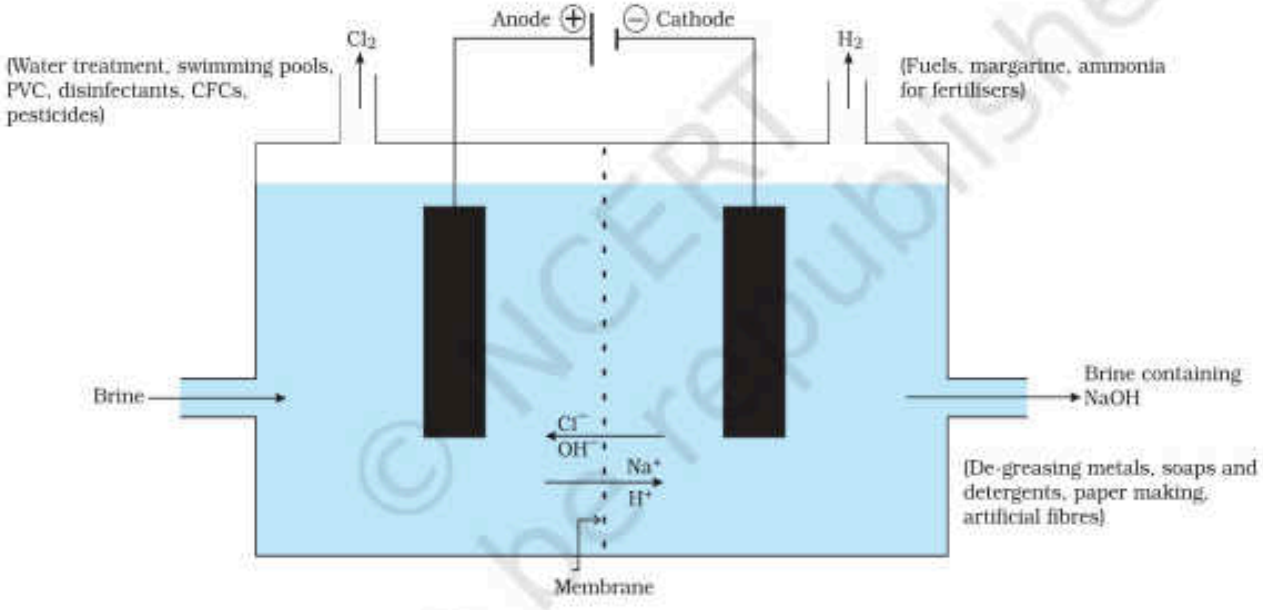


Figure 2.8 Important products from the chlor-alkali process

Bleaching powder

You have already come to know that chlorine is produced during the electrolysis of aqueous sodium chloride (brine). This chlorine gas is used for the manufacture of bleaching powder. Bleaching powder is produced by the action of chlorine on dry slaked lime [$Ca(OH)_2$]. Bleaching powder is represented as $Ca(ClO)_2$, though the actual composition is quite complex.

$$2Ca(OH)_2 + 2Cl_2 \rightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$$

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Bleaching powder is used –

(i) for bleaching cotton and linen in the textile industry, for bleaching wood pulp in paper factories and for bleaching washed clothes in laundry;
(ii) as an oxidising agent in many chemical industries; and
(iii) to make drinking water free from germs.

Baking soda

The baking soda is commonly used in the kitchen for making tasty crispy pakoras, etc. Sometimes it is added for faster cooking. The chemical name of the compound is sodium hydrogencarbonate ($NaHCO_3$). It is produced using sodium chloride as one of the raw materials.

$$NaCl + H_2O + CO_2 + NH_3 \rightarrow \underset{\text{(Ammonium chloride)}}{NH_4Cl} + \underset{\text{(Sodium hydrogencarbonate)}}{NaHCO_3}$$

Did you check the pH of sodium hydrogencarbonate in Activity 2.14? Can you correlate why it can be used to neutralise an acid? It is a mild non-corrosive basic salt. The following reaction takes place when it is heated during cooking –

$$\underset{\text{(Sodium hydrogencarbonate)}}{2NaHCO_3} \xrightarrow{\text{Heat}} \underset{\text{(Sodium carbonate)}}{Na_2CO_3} + H_2O + CO_2$$

Sodium hydrogencarbonate has got various uses in the household.

Uses of Baking soda

(i) Baking powder is a mixture of baking soda (sodium hydrogen carbonate) and tartaric acid which is a weak edible acid. When baking powder is heated or mixed in water, the following reaction takes place –

$$NaHCO_3 + \underset{\text{(From any acid)}}{H^+} \rightarrow CO_2 + H_2O + \text{Sodium salt of acid}$$

Carbon dioxide produced during the reaction can cause bread or cake to rise making them soft and spongy.

(ii) Sodium hydrogencarbonate is also an ingredient in antacids. Being alkaline, it neutralises excess acid in the stomach and provides relief.
(iii) It is also used in soda-acid fire extinguishers.

Washing soda

Another chemical that can be obtained from sodium chloride is $Na_2CO_3.10H_2O$ (washing soda). You have seen above that sodium carbonate can be obtained by heating baking soda; recrystallisation of sodium carbonate gives washing soda. It is also a basic salt.

$$\underset{\text{(Sodium carbonate)}}{Na_2CO_3} + 10H_2O \rightarrow Na_2CO_3.10H_2O$$

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What does $10H_2O$ signify? Does it make Na_2CO_3 wet? We will address this question in the next section.

Sodium carbonate and sodium hydrogencarbonate are useful chemicals for many industrial processes as well.

Uses of washing soda

(i) Sodium carbonate (washing soda) is used in glass, soap and paper industries.
(ii) It is used in the manufacture of sodium compounds such as borax.
(iii) Sodium carbonate can be used as a cleaning agent for domestic purposes.
(iv) It is used for removing permanent hardness of water.

2.4.4 Are the Crystals of Salts really Dry?

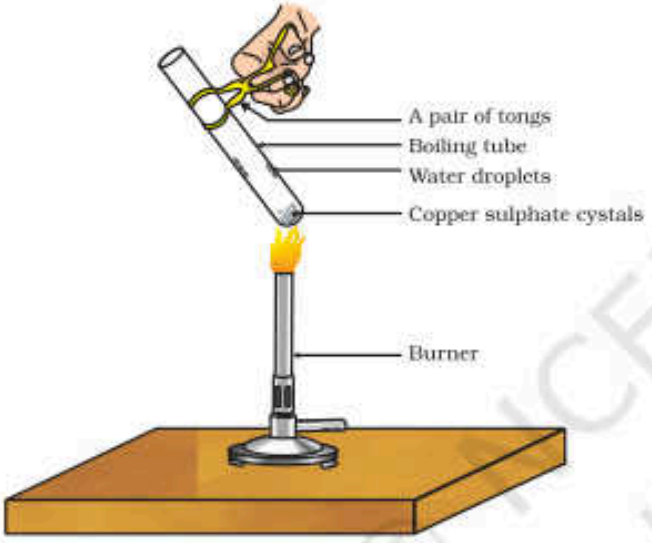


Figure 2.9
Removing water of crystallisation

Activity 2.15

- Heat a few crystals of copper sulphate in a dry boiling tube.
- What is the colour of the copper sulphate after heating?
- Do you notice water droplets in the boiling tube? Where have these come from?
- Add 2-3 drops of water on the sample of copper sulphate obtained after heating.
- What do you observe? Is the blue colour of copper sulphate restored?

Copper sulphate crystals which seem to be dry contain water of crystallisation. When we heat the crystals, this water is removed and the salt turns white.

If you moisten the crystals again with water, you will find that blue colour of the crystals reappears.

Water of crystallisation is the fixed number of water molecules present in one formula unit of a salt. Five water molecules are present in one formula unit of copper sulphate. Chemical formula for hydrated copper sulphate is $Cu\,SO_4.\,5H_2O$. Now you would be able to answer the question whether the molecule of $Na_2CO_3.10H_2O$ is wet.

One other salt, which possesses water of crystallisation is gypsum. It has two water molecules as water of cyrstallisation. It has the chemical formula $CaSO_4.2H_2O$. Let us look into the use of this salt.

Plaster of Paris

On heating gypsum at 373 K, it loses water molecules and becomes calcium sulphate hemihydrate ($CaSO_4 \cdot \frac{1}{2} H_2O$). This is called Plaster of

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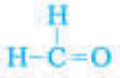
4. Draw the structures for the following compounds.

(i) Ethanoic acid (ii) Bromopentane*

(iii) Butanone (iv) Hexanal.

*Are structural isomers possible for bromopentane?

5. How would you name the following compounds?

(i) $CH_3—CH_2—Br$ (ii) 

(iii) 

4.3 CHEMICAL PROPERTIES OF CARBON COMPOUNDS

In this section we will be studying about some of the chemical properties of carbon compounds. Since most of the fuels we use are either carbon or its compounds, we shall first study combustion.

4.3.1 Combustion

Carbon, in all its allotropic forms, burns in oxygen to give carbon dioxide along with the release of heat and light. Most carbon compounds also release a large amount of heat and light on burning. These are the oxidation reactions that you learnt about in the first Chapter –

(i) $C + O_2 \rightarrow CO_2$ + heat and light

(ii) $CH_4 + O_2 \rightarrow CO_2 + H_2O$ + heat and light

(iii) $CH_3CH_2OH + O_2 \rightarrow CO_2 + H_2O$ + heat and light

Balance the latter two reactions like you learnt in the first Chapter.

Activity 4.3

CAUTION: This Activity needs the teacher's assistance.

- Take some carbon compounds (naphthalene, camphor, alcohol) one by one on a spatula and burn them.
- Observe the nature of the flame and note whether smoke is produced.
- Place a metal plate above the flame. Is there a deposition on the plate in case of any of the compounds?

Activity 4.4

- Light a bunsen burner and adjust the air hole at the base to get different types of flames/presence of smoke.
- When do you get a yellow, sooty flame?
- When do you get a blue flame?

Saturated hydrocarbons will generally give a clean flame while unsaturated carbon compounds will give a yellow flame with lots of black smoke. This results in a sooty deposit on the metal plate in Activity 4.3. However, limiting the supply of air results in incomplete combustion of even saturated hydrocarbons giving a sooty flame. The gas/kerosene stove used at home has inlets for air so that a sufficiently oxygen-rich

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mixture is burnt to give a clean blue flame. If you observe the bottoms of cooking vessels getting blackened, it means that the air holes are blocked and fuel is getting wasted. Fuels such as coal and petroleum have some amount of nitrogen and sulphur in them. Their combustion results in the formation of oxides of sulphur and nitrogen which are major pollutants in the environment.

Do You Know?

Why do substances burn with or without a flame?

Have you ever observed either a coal or a wood fire? If not, the next time you get a chance, take close note of what happens when the wood or coal starts to burn. You have seen above that a candle or the LPG in the gas stove burns with a flame. However, you will observe the coal or charcoal in an 'angithi' sometimes just glows red and gives out heat without a flame. This is because a flame is only produced when gaseous substances burn. When wood or charcoal is ignited, the volatile substances present vapourise and burn with a flame in the beginning.

A luminous flame is seen when the atoms of the gaseous substance are heated and start to glow. The colour produced by each element is a characteristic property of that element. Try and heat a copper wire in the flame of a gas stove and observe its colour. You have seen that incomplete combustion gives soot which is carbon. On this basis, what will you attribute the yellow colour of a candle flame to?

More to Know!

Formation of coal and petroleum

Coal and petroleum have been formed from biomass which has been subjected to various biological and geological processes. Coal is the remains of trees, ferns, and other plants that lived millions of years ago. These were crushed into the earth, perhaps by earthquakes or volcanic eruptions. They were pressed down by layers of earth and rock. They slowly decayed into coal. Oil and gas are the remains of millions of tiny plants and animals that lived in the sea. When they died, their bodies sank to the sea bed and were covered by silt. Bacteria attacked the dead remains, turning them into oil and gas under the high pressures they were being subjected to. Meanwhile, the silt was slowly compressed into rock. The oil and gas seeped into the porous parts of the rock, and got trapped like water in a sponge. Can you guess why coal and petroleum are called fossil fuels?

4.3.2 Oxidation

Activity 4.5

- Take about 3 mL of ethanol in a test tube and warm it gently in a water bath.
- Add a 5% solution of alkaline potassium permanganate drop by drop to this solution.
- Does the colour of potassium permanganate persist when it is added initially?
- Why does the colour of potassium permanganate not disappear when excess is added?

You have learnt about oxidation reactions in the first Chapter. Carbon compounds can be easily oxidised on combustion. In addition to this complete oxidation, we have reactions in which alcohols are converted to carboxylic acids –

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Activity 5.3

- Take 1 mL starch solution (1%) in two test tubes (A and B).
- Add 1 mL saliva to test tube A and leave both test tubes undisturbed for 20-30 minutes.
- Now add a few drops of dilute iodine solution to the test tubes.
- In which test tube do you observe a colour change?
- What does this indicate about the presence or absence of starch in the two test tubes?
- What does this tell us about the action of saliva on starch?

We eat various types of food which has to pass through the same digestive tract. Naturally the food has to be processed to generate particles which are small and of the same texture. This is achieved by crushing the food with our teeth. Since the lining of the canal is soft, the food is also wetted to make its passage smooth. When we eat something we like, our mouth 'waters'. This is actually not only water, but a fluid called saliva secreted by the salivary glands. Another aspect of the food we ingest is its complex nature. If it is to be absorbed from the alimentary canal, it has to be broken into smaller molecules. This is done with the help of biological catalysts called enzymes. The saliva contains an enzyme called salivary amylase that breaks down starch which is a complex molecule to give simple sugar. The food is mixed thoroughly with saliva and moved around the mouth while chewing by the muscular tongue.

It is necessary to move the food in a regulated manner along the digestive tube so that it can be processed properly in each part. The lining of canal has muscles that contract rhythmically in order to push the food forward. These peristaltic movements occur all along the gut.

From the mouth, the food is taken to the stomach through the food-pipe or oesophagus. The stomach is a large organ which expands when food enters it. The muscular walls of the stomach help in mixing the food thoroughly with more digestive juices.

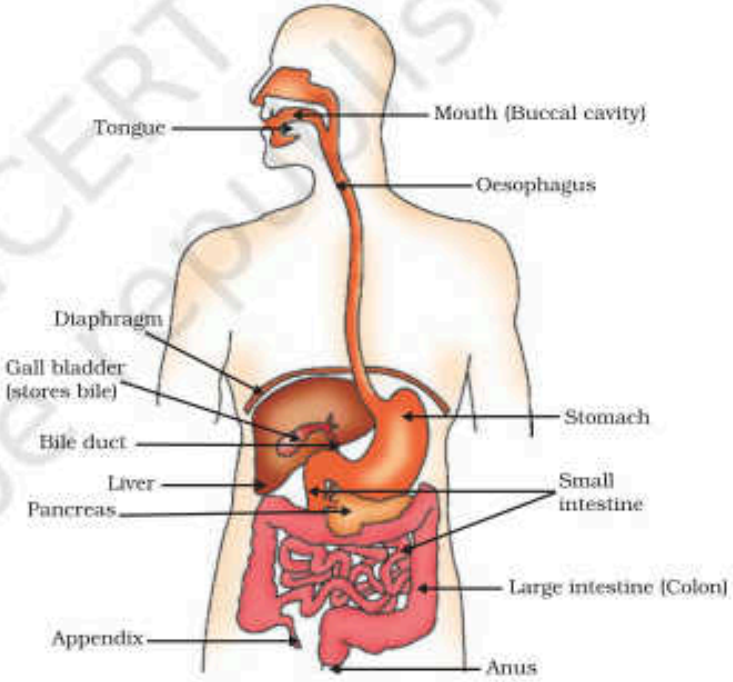


Figure 5.6 *Human alimentary canal*

The digestion in stomach is taken care of by the gastric glands present in the wall of the stomach. These release hydrochloric acid, a protein digesting enzyme called pepsin, and mucus. The hydrochloric acid creates an acidic medium which facilitates the action of the enzyme pepsin. What other function do you think is served by the acid? The mucus protects the inner lining of the stomach from the action of the acid under normal conditions. We

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have often heard adults complaining about 'acidity'. Can this be related to what has been discussed above?

The exit of food from the stomach is regulated by a sphincter muscle which releases it in small amounts into the small intestine. From the stomach, the food now enters the small intestine. This is the longest part of the alimentary canal which is fitted into a compact space because of extensive coiling. The length of the small intestine differs in various animals depending on the food they eat. Herbivores eating grass need a longer small intestine to allow the cellulose to be digested. Meat is easier to digest, hence carnivores like tigers have a shorter small intestine.

The small intestine is the site of the complete digestion of carbohydrates, proteins and fats. It receives the secretions of the liver and pancreas for this purpose. The food coming from the stomach is acidic and has to be made alkaline for the pancreatic enzymes to act. Bile juice from the liver accomplishes this in addition to acting on fats. Fats are present in the intestine in the form of large globules which makes it difficult for enzymes to act on them. Bile salts break them down into smaller globules increasing the efficiency of enzyme action. This is similar to the emulsifying action of soaps on dirt that we have learnt about in Chapter 4. The pancreas secretes pancreatic juice which contains enzymes like trypsin for digesting proteins and lipase for breaking down emulsified fats. The walls of the small intestine contain glands which secrete intestinal juice. The enzymes present in it finally convert the proteins to amino acids, complex carbohydrates into glucose and fats into fatty acids and glycerol.

Digested food is taken up by the walls of the intestine. The inner lining of the small intestine has numerous finger-like projections called villi which increase the surface area for absorption. The villi are richly supplied with blood vessels which take the absorbed food to each and every cell of the body, where it is utilised for obtaining energy, building up new tissues and the repair of old tissues.

The unabsorbed food is sent into the large intestine where its wall absorb more water from this material. The rest of the material is removed from the body via the anus. The exit of this waste material is regulated by the anal sphincter.

More to Know!

Dental caries

Dental caries or tooth decay causes gradual softening of enamel and dentine. It begins when bacteria acting on sugars produce acids that softens or demineralises the enamel. Masses of bacterial cells together with food particles stick to the teeth to form dental plaque. Saliva cannot reach the tooth surface to neutralise the acid as plaque covers the teeth. Brushing the teeth after eating removes the plaque before the bacteria produce acids. If untreated, microorganisms may invade the pulp, causing inflammation and infection.

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QUESTIONS

1. What are the differences between autotrophic nutrition and heterotrophic nutrition?
2. Where do plants get each of the raw materials required for photosynthesis?
3. What is the role of the acid in our stomach?
4. What is the function of digestive enzymes?
5. How is the small intestine designed to absorb digested food?

5.3 RESPIRATION

Activity 5.4

- Take some freshly prepared lime water in a test tube.
- Blow air through this lime water.
- Note how long it takes for the lime water to turn milky.
- Use a syringe or *pichkari* to pass air through some fresh lime water taken in another test tube (Fig. 5.7).
- Note how long it takes for this lime water to turn milky.
- What does this tell us about the amount of carbon dioxide in the air that we breathe out?

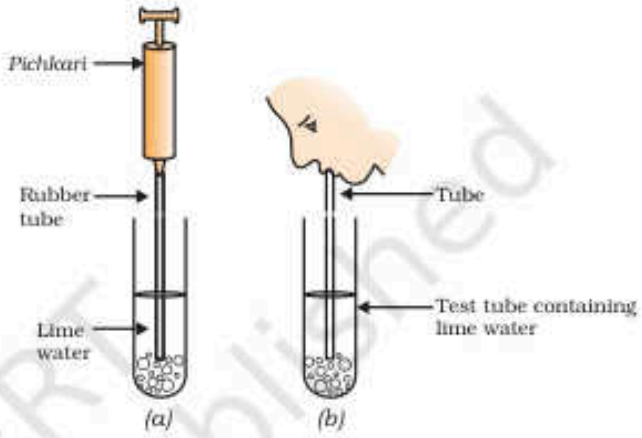


Figure 5.7
(a) Air being passed into lime water with a pichkari/ syringe, (b) air being exhaled into lime water

Activity 5.5

- Take some fruit juice or sugar solution and add some yeast to this. Take this mixture in a test tube fitted with a one-holed cork.
- Fit the cork with a bent glass tube. Dip the free end of the glass tube into a test tube containing freshly prepared lime water.
- What change is observed in the lime water and how long does it take for this change to occur?
- What does this tell us about the products of fermentation?

We have discussed nutrition in organisms in the last section. The food material taken in during the process of nutrition is used in cells to provide energy for various life processes. Diverse organisms do this in different ways – some use oxygen to break-down glucose completely into carbon dioxide and water, some use other pathways that do not involve oxygen (Fig. 5.8). In all cases, the first step is the break-down of glucose, a six-carbon molecule, into a three-carbon molecule called pyruvate. This process takes place in the cytoplasm. Further, the pyruvate may be converted into ethanol and carbon dioxide. This process takes place in yeast during fermentation. Since this process takes place in the absence of air (oxygen), it is called anaerobic respiration. Break-down of pyruvate using oxygen takes place in the mitochondria. This

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process breaks up the three-carbon pyruvate molecule to give three molecules of carbon dioxide. The other product is water. Since this process takes place in the presence of air (oxygen), it is called aerobic respiration. The release of energy in this aerobic process is a lot greater than in the anaerobic process. Sometimes, when there is a lack of oxygen in our muscle cells, another pathway for the break-down of pyruvate is taken. Here the pyruvate is converted into lactic acid which is also a three-carbon molecule. This build-up of lactic acid in our muscles during sudden activity causes cramps.

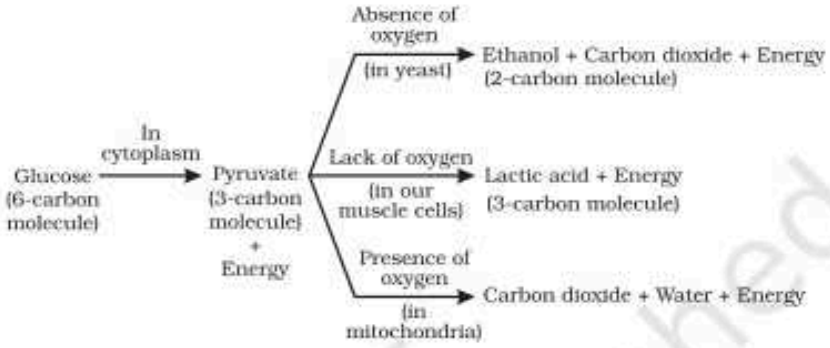


Figure 5.8 *Break-down of glucose by various pathways*

The energy released during cellular respiration is immediately used to synthesise a molecule called ATP which is used to fuel all other activities in the cell. In these processes, ATP is broken down giving rise to a fixed amount of energy which can drive the endothermic reactions taking place in the cell.

More to Know!

ATP

ATP is the energy currency for most cellular processes. The energy released during the process of respiration is used to make an ATP molecule from ADP and inorganic phosphate.

$$ADP + Ⓟ \xrightarrow{\text{Energy}} ADP\sim Ⓟ = ATP$$

Ⓟ: Phosphate

Endothermic processes in the cell then use this ATP to drive the reactions. When the terminal phosphate linkage in ATP is broken using water, the energy equivalent to 30.5 kJ/mol is released.

Think of how a battery can provide energy for many different kinds of uses. It can be used to obtain mechanical energy, light energy, electrical energy and so on. Similarly, ATP can be used in the cells for the contraction of muscles, protein synthesis, conduction of nervous impulses and many other activities.

Since the aerobic respiration pathway depends on oxygen, aerobic organisms need to ensure that there is sufficient intake of oxygen. We have seen that plants exchange gases through stomata, and the large inter-cellular spaces ensure that all cells are in contact with air. Carbon dioxide and oxygen are exchanged by diffusion here. They can go into

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cells, or away from them and out into the air. The direction of diffusion depends upon the environmental conditions and the requirements of the plant. At night, when there is no photosynthesis occurring, CO_2 elimination is the major exchange activity going on. During the day, CO_2 generated during respiration is used up for photosynthesis, hence there is no CO_2 release. Instead, oxygen release is the major event at this time.

Animals have evolved different organs for the uptake of oxygen from the environment and for getting rid of the carbon dioxide produced. Terrestrial animals can breathe the oxygen in the atmosphere, but animals that live in water need to use the oxygen dissolved in water.

Activity 5.6

- Observe fish in an aquarium. They open and close their mouths and the gill-slits (or the operculum which covers the gill-slits) behind their eyes also open and close. Are the timings of the opening and closing of the mouth and gill-slits coordinated in some manner?
- Count the number of times the fish opens and closes its mouth in a minute.
- Compare this to the number of times you breathe in and out in a minute.

Since the amount of dissolved oxygen is fairly low compared to the amount of oxygen in the air, the rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms. Fishes take in water through their mouths and force it past the gills where the dissolved oxygen is taken up by blood.

Terrestrial organisms use the oxygen in the atmosphere for respiration. This oxygen is absorbed by different organs in different animals. All these organs have a structure that increases the surface area which is in contact with the oxygen-rich atmosphere. Since the exchange of oxygen and carbon dioxide has to take place across this surface, this surface is very fine and delicate. In order to protect this surface, it is usually placed within the body, so there have to be passages that will take air to this area. In addition, there is a mechanism for moving the air in and out of this area where the oxygen is absorbed.

In human beings (Fig. 5.9), air is taken into the body through the nostrils. The air passing through the nostrils is filtered by fine hairs that line the passage. The passage is also lined with mucus which helps in this process. From here, the air passes through the throat and into the lungs. Rings of cartilage are present in the throat. These ensure that the air-passage does not collapse.

More to Know!

Using tobacco directly or any product of tobacco in the form of cigar, cigarettes, *bidis*, *hookah*, *gutkha*, etc., is harmful. Use of tobacco most commonly affects the tongue, lungs, heart and liver. Smokeless tobacco is also a major risk factor for heart attacks, strokes, pulmonary diseases and several forms of cancers. There is a high incidence of oral cancer in India due to the chewing of tobacco in the form of *gutkha*. Stay healthy; just say NO to tobacco and its products!

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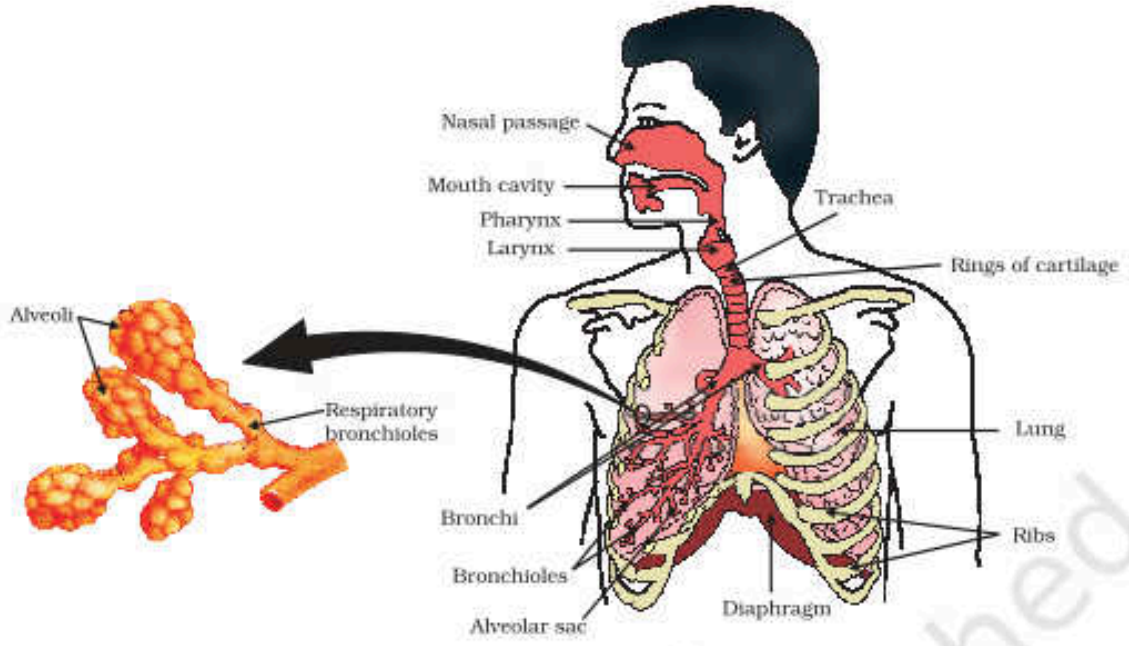


Figure 5.9 *Human respiratory system*

Do You Know?

Smoking is injurious to health.

Lung cancer is one of common causes of deaths in the world. The upper part of respiratory tract is provided with small hair-like structures called cilia. These cilia help to remove germs, dust and other harmful particles from inhaled air. Smoking destroys these hair due to which germs, dust, smoke and other harmful chemicals enter lungs and cause infection, cough and even lung cancer.

Within the lungs, the passage divides into smaller and smaller tubes which finally terminate in balloon-like structures which are called alveoli (singular–alveolus). The alveoli provide a surface where the exchange of gases can take place. The walls of the alveoli contain an extensive network of blood-vessels. As we have seen in earlier years, when we breathe in, we lift our ribs and flatten our diaphragm, and the chest cavity becomes larger as a result. Because of this, air is sucked into the lungs and fills the expanded alveoli. The blood brings carbon dioxide from the rest of the body for release into the alveoli, and the oxygen in the alveolar air is taken up by blood in the alveolar blood vessels to be transported to all the cells in the body. During the breathing cycle, when air is taken in and let out, the lungs always contain a residual volume of air so that there is sufficient time for oxygen to be absorbed and for the carbon dioxide to be released.

When the body size of animals is large, the diffusion pressure alone cannot take care of oxygen delivery to all parts of the body. Instead, respiratory pigments take up oxygen from the air in the lungs and carry it to tissues which are deficient in oxygen before releasing it. In human beings, the respiratory pigment is haemoglobin which has a very high affinity for oxygen. This pigment is present in the red blood corpuscles. Carbon dioxide is more soluble in water than oxygen is and hence is mostly transported in the dissolved form in our blood.

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Do You Know?

- If the alveolar surface were spread out, it would cover about 80 m^2. How much do you think the surface area of your body is? Consider how efficient exchange of gases becomes because of the large surface available for the exchange to take place.
- If diffusion were to move oxygen in our body, it is estimated that it would take 3 years for a molecule of oxygen to get to our toes from our lungs. Aren't you glad that we have haemoglobin?

QUESTIONS

1. What advantage over an aquatic organism does a terrestrial organism have with regard to obtaining oxygen for respiration?
2. What are the different ways in which glucose is oxidised to provide energy in various organisms?
3. How is oxygen and carbon dioxide transported in human beings?
4. How are the lungs designed in human beings to maximise the area for exchange of gases?

5.4 TRANSPORTATION

5.4.1 Transportation in Human Beings

Activity 5.7

- Visit a health centre in your locality and find out what is the normal range of haemoglobin content in human beings.
- Is it the same for children and adults?
- Is there any difference in the haemoglobin levels for men and women?
- Visit a veterinary clinic in your locality. Find out what is the normal range of haemoglobin content in an animal like the buffalo or cow.
- Is this content different in calves, male and female animals?
- Compare the difference seen in male and female human beings and animals.
- How would the difference, if any, be explained?

We have seen in previous sections that blood transports food, oxygen and waste materials in our bodies. In Class IX, we learnt about blood being a fluid connective tissue. Blood consists of a fluid medium called plasma in which the cells are suspended. Plasma transports food, carbon dioxide and nitrogenous wastes in dissolved form. Oxygen is carried by the red blood corpuscles. Many other substances like salts, are also transported by the blood. We thus need a pumping organ to push blood around the body, a network of tubes to reach all the tissues and a system in place to ensure that this network can be repaired if damaged.

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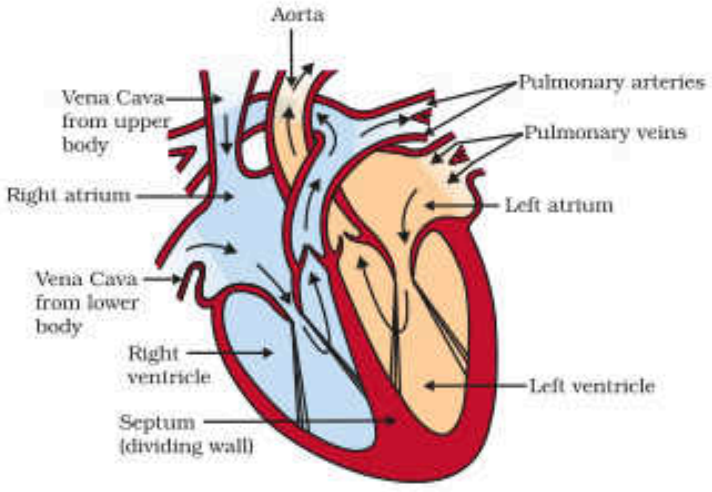


Figure 5.10
Schematic sectional view of the human heart

Our pump — the heart

The heart is a muscular organ which is as big as our fist (Fig. 5.10). Because both oxygen and carbon dioxide have to be transported by the blood, the heart has different chambers to prevent the oxygen-rich blood from mixing with the blood containing carbon dioxide. The carbon dioxide-rich blood has to reach the lungs for the carbon dioxide to be removed, and the oxygenated blood from the lungs has to be brought back to the heart. This oxygen-rich blood is then pumped to the rest of the body.

We can follow this process step by step (Fig. 5.11). Oxygen-rich blood from the lungs comes to the thin-walled upper chamber of the heart on the left, the left atrium. The left atrium relaxes when it is collecting this blood. It then contracts, while the next chamber, the left ventricle, relaxes, so that the blood is transferred to it. When the muscular left ventricle contracts in its turn, the blood is pumped out to the body. De-oxygenated blood comes from the body to the upper chamber on the right, the right atrium, as it relaxes. As the right atrium contracts, the corresponding lower chamber, the right ventricle, dilates. This transfers blood to the right ventricle, which in turn pumps it to the lungs for oxygenation. Since ventricles have to pump blood into various organs, they have thicker muscular walls than the atria do. Valves ensure that blood does not flow backwards when the atria or ventricles contract.

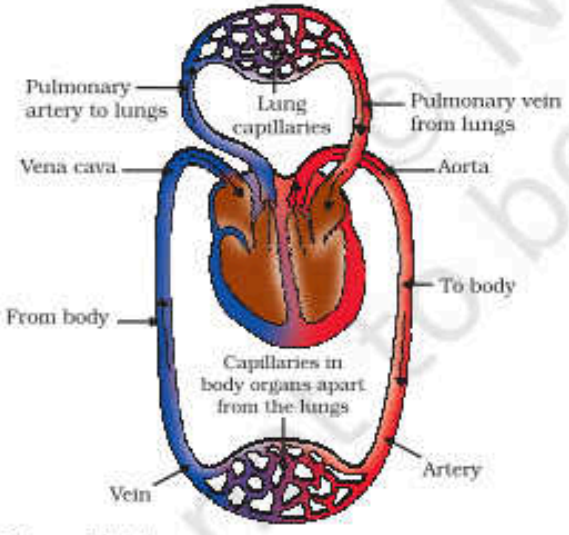


Figure 5.11
Schematic representation of transport and exchange of oxygen and carbon dioxide

Oxygen enters the blood in the lungs

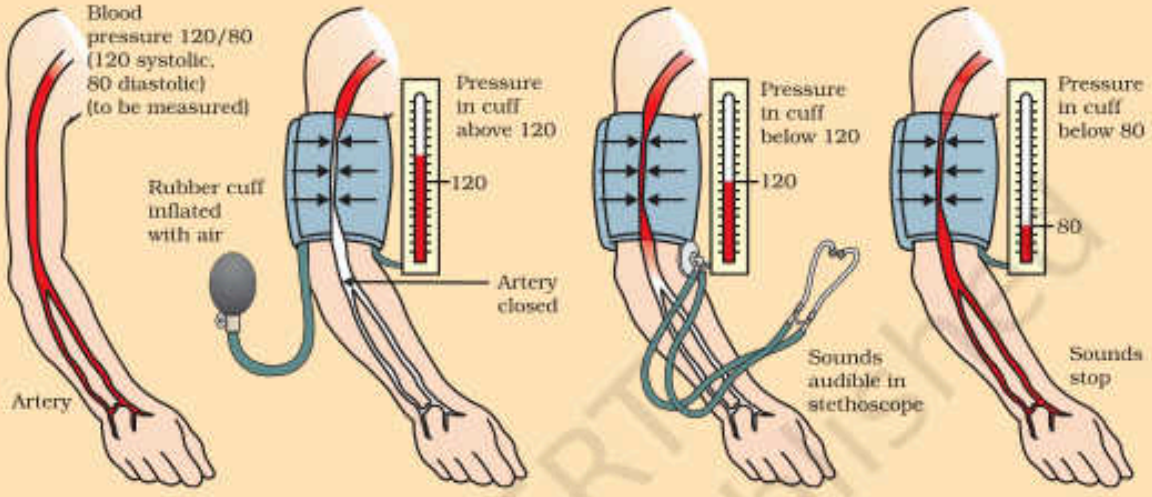
The separation of the right side and the left side of the heart is useful to keep oxygenated and de-oxygenated blood from mixing. Such separation allows a highly efficient supply of oxygen to the body. This is useful in animals that have high energy needs, such as birds and mammals, which constantly use energy to maintain their body temperature. In animals that do not use energy for this purpose, the body temperature depends on the temperature in the environment. Such animals, like amphibians or many reptiles have three-chambered hearts, and tolerate some mixing of the oxygenated and de-oxygenated blood streams. Fishes, on the other hand, have only two chambers to their hearts, and the blood is pumped to the gills, is oxygenated there, and passes directly to the rest of the body. Thus, blood goes only once through the heart in the fish during one cycle of passage through the body. On the other hand, it goes through the heart twice during each cycle in other vertebrates. This is known as double circulation.

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More to Know!

Blood pressure

The force that blood exerts against the wall of a vessel is called blood pressure. This pressure is much greater in arteries than in veins. The pressure of blood inside the artery during ventricular systole (contraction) is called systolic pressure and pressure in artery during ventricular diastole (relaxation) is called diastolic pressure. The normal systolic pressure is about 120 mm of Hg and diastolic pressure is 80 mm of Hg.



Blood pressure is measured with an instrument called sphygmomanometer. High blood pressure is also called hypertension and is caused by the constriction of arterioles, which results in increased resistance to blood flow. It can lead to the rupture of an artery and internal bleeding.

The tubes – blood vessels

Arteries are the vessels which carry blood away from the heart to various organs of the body. Since the blood emerges from the heart under high pressure, the arteries have thick, elastic walls. Veins collect the blood from different organs and bring it back to the heart. They do not need thick walls because the blood is no longer under pressure, instead they have valves that ensure that the blood flows only in one direction.

On reaching an organ or tissue, the artery divides into smaller and smaller vessels to bring the blood in contact with all the individual cells. The smallest vessels have walls which are one-cell thick and are called capillaries. Exchange of material between the blood and surrounding cells takes place across this thin wall. The capillaries then join together to form veins that convey the blood away from the organ or tissue.

Maintenance by platelets

What happens if this system of tubes develops a leak? Think about situations when we are injured and start bleeding. Naturally the loss of blood from the system has to be minimised. In addition, leakage would lead to a loss of pressure which would reduce the efficiency of the

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and animals. This is one of the ways in which they enter the food chain. As these chemicals are not degradable, these get accumulated progressively at each trophic level. As human beings occupy the top level in any food chain, the maximum concentration of these chemicals get accumulated in our bodies. This phenomenon is known as biological magnification. This is the reason why our food grains such as wheat and rice, vegetables and fruits, and even meat, contain varying amounts of pesticide residues. They cannot always be removed by washing or other means.

Activity 13.3

- Newspaper reports about pesticide levels in ready-made food items are often seen these days and some states have banned these products. Debate in groups the need for such bans.
- What do you think would be the source of pesticides in these food items? Could pesticides get into our bodies from this source through other food products too?
- Discuss what methods could be applied to reduce our intake of pesticides.

QUESTIONS

1. What are trophic levels? Give an example of a food chain and state the different trophic levels in it.
2. What is the role of decomposers in the ecosystem?

13.2 HOW DO OUR ACTIVITIES AFFECT THE ENVIRONMENT?

We are an integral part of the environment. Changes in the environment affect us and our activities change the environment around us. We have already seen in Class IX how our activities pollute the environment. In this chapter, we shall be looking at two of the environmental problems in detail, that is, depletion of the ozone layer and waste disposal.

13.2.1 Ozone Layer and How it is Getting Depleted

Ozone (O_3) is a molecule formed by three atoms of oxygen. While O_2, which we normally refer to as oxygen, is essential for all aerobic forms of life, Ozone, is a deadly poison. However, at the higher levels of the atmosphere, ozone performs an essential function. It shields the surface of the earth from ultraviolet (UV) radiation from the Sun. This radiation

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is highly damaging to organisms, for example, it is known to cause skin cancer in human beings.

Ozone at the higher levels of the atmosphere is a product of UV radiation acting on oxygen (O_2) molecule. The higher energy UV radiations split apart some moleculer oxygen (O_2) into free oxygen (O) atoms. These atoms then combine with the molecular oxygen to form ozone as shown—

$$O_2 \xrightarrow{UV} O + O$$

$$O + O_2 \rightarrow \underset{\text{(Ozone)}}{O_3}$$

The amount of ozone in the atmosphere began to drop sharply in the 1980s. This decrease has been linked to synthetic chemicals like chlorofluorocarbons (CFCs) which are used as refrigerants and in fire extinguishers. In 1987, the United Nations Environment Programme (UNEP) succeeded in forging an agreement to freeze CFC production at 1986 levels. It is now mandatory for all the manufacturing companies to make CFC-free refrigerators throughout the world.

Activity 13.4

- Find out from the library, internet or newspaper reports, which chemicals are responsible for the depletion of the ozone layer.
- Find out if the regulations put in place to control the emission of these chemicals have succeeded in reducing the damage to the ozone layer. Has the size of the hole in the ozone layer changed in recent years?

13.2.2 Managing the Garbage we Produce

In our daily activities, we generate a lot of material that are thrown away. What are some of these waste materials? What happens after we throw them away? Let us perform an activity to find answers to these questions.

Activity 13.5

- Collect waste material from your homes. This could include all the waste generated during a day, like kitchen waste (spoilt food, vegetable peels, used tea leaves, milk packets and empty cartons), waste paper, empty medicine bottles/strips/bubble packs, old and torn clothes and broken footwear.
- Bury this material in a pit in the school garden or if there is no space available, you can collect the material in an old bucket/ flower pot and cover with at least 15 cm of soil.
- Keep this material moist and observe at 15-day intervals.
- What are the materials that remain unchanged over long periods of time?
- What are the materials which change their form and structure over time?
- Of these materials that are changed, which ones change the fastest?

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We have seen in the chapter on 'Life Processes' that the food we eat is digested by various enzymes in our body. Have you ever wondered why the same enzyme does not break-down everything we eat? Enzymes are specific in their action, specific enzymes are needed for the break-down of a particular substance. That is why we will not get any energy if we try to eat coal! Because of this, many human-made materials like plastics will not be broken down by the action of bacteria or other saprophytes. These materials will be acted upon by physical processes like heat and pressure, but under the ambient conditions found in our environment, these persist for a long time.

Substances that are broken down by biological processes are said to be biodegradable. How many of the substances you buried were biodegradable? Substances that are not broken down in this manner are said to be non-biodegradable. These substances may be inert and simply persist in the environment for a long time or may harm the various members of the eco-system.

Activity 13.6

- Use the library or internet to find out more about biodegradable and non-biodegradable substances.
- How long are various non-biodegradable substances expected to last in our environment?
- These days, new types of plastics which are said to be biodegradable are available. Find out more about such materials and whether they do or do not harm the environment.

QUESTIONS

1. Why are some substances biodegradable and some non-biodegradable?
2. Give any two ways in which biodegradable substances would affect the environment.
3. Give any two ways in which non-biodegradable substances would affect the environment.

Visit any town or city, and we are sure to find heaps of garbage all over the place. Visit any place of tourist interest and we are sure to find the place littered with empty food wrappers. In the earlier classes we have talked about this problem of dealing with the garbage that we generate. Let us now look at the problem a bit more deeply.

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Activity 13.7

- Find out what happens to the waste generated at home. Is there a system in place to collect this waste?
- Find out how the local body (*panchayat*, municipal corporation, resident welfare association) deals with the waste. Are there mechanisms in place to treat the biodegradable and non-biodegradable wastes separately?
- Calculate how much waste is generated at home in a day.
- How much of this waste is biodegradable?
- Calculate how much waste is generated in the classroom in a day.
- How much of this waste is biodegradable?
- Suggest ways of dealing with this waste.

Activity 13.8

- Find out how the sewage in your locality is treated. Are there mechanisms in place to ensure that local water bodies are not polluted by untreated sewage.
- Find out how the local industries in your locality treat their wastes. Are there mechanisms in place to ensure that the soil and water are not polluted by this waste?

Improvements in our life-style have resulted in greater amounts of waste material generation. Changes in attitude also have a role to play, with more and more things we use becoming disposable. Changes in packaging have resulted in much of our waste becoming non-biodegradable. What do you think will be the impact of these on our environment?

Think it over

Disposable cups in trains

If you ask your parents, they will probably remember a time when tea in trains was served in plastic glasses which had to be returned to the vendor. The introduction of disposable cups was hailed as a step forward for reasons of hygiene. No one at that time perhaps thought about the impact caused by the disposal of millions of these cups on a daily basis. Some time back, *kulhads*, that is, disposable cups made of clay, were suggested as an alternative. But a little thought showed that making these *kulhads* on a large scale would result in the loss of the fertile top-soil. Now disposable paper-cups are being used. What do you think are the advantages of disposable paper-cups over disposable plastic cups?

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fringes of these forests depends upon it. A part of the land is termed as waste land and land put to other non-agricultural uses. Waste land includes rocky, arid and desert areas and land put to other non-agricultural uses includes settlements, roads, railways, industry etc. Continuous use of land over a long period of time without taking appropriate measures to conserve and manage it, has resulted in land degradation. This, in turn, has serious repercussions on society and the environment.

Land Degradation and Conservation Measures

We have shared our land with the past generations and will have to do so with the future generations too. Ninety-five per cent of our basic needs for food, shelter and clothing are obtained from land. Human activities have not only brought about degradation of land but have also aggravated the pace of natural forces to cause damage to land.

Some human activities such as deforestation, over grazing, mining and quarrying too have contributed significantly in land degradation.

Mining sites are abandoned after excavation work is complete leaving deep scars and traces of over-burdening. In states like Jharkhand, Chhattisgarh, Madhya Pradesh and Odisha deforestation due to mining have caused severe land degradation. In states like Gujarat, Rajasthan, Madhya Pradesh and Maharashtra overgrazing is one of the main reasons for land degradation. In the states of Punjab, Haryana, western Uttar Pradesh, over irrigation is responsible for land degradation due to water logging leading to increase in salinity and alkalinity in the soil. The mineral processing like grinding of limestone for cement industry and calcite and soapstone for ceramic industry generate huge quantity of dust in the atmosphere. It retards the process of infiltration of water into the soil after it settles down on the land. In recent years, industrial effluents as waste have become a major source of land and water pollution in many parts of the country.

There are many ways to solve the problems of land degradation. Afforestation and proper management of grazing can help to some extent. Planting of shelter belts of plants, control on over grazing, stabilisation of sand dunes by growing thorny bushes are some of the methods to check land degradation in arid areas. Proper management of waste lands, control of mining activities, proper discharge and disposal of industrial effluents and wastes after treatment can reduce land and water degradation in industrial and suburban areas.

Soil as a Resource

Soil is the most important natural resource. It is the medium of plant growth and supports different types of living organisms on the earth. The soil is a living system. It takes millions of years to form soil upto a few cm in depth. Relief, parent rock or bed rock, climate, vegetation and other forms of life and time are important factors in the formation of soil. Various forces of nature such as change in temperature, actions of running water, wind and glaciers, activities of decomposers etc. contribute to the formation of soil. Chemical and organic changes which take place in the



Fig. 1.5: Soil Profile

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soil are equally important. Soil also consists of organic (humus) and inorganic materials (Fig. 1.5).

On the basis of the factors responsible for soil formation, colour, thickness, texture, age, chemical and physical properties, the soils of India are classified in different types.

Classification of Soils

India has varied relief features, landforms, climatic realms and vegetation types. These have contributed in the development of various types of soils.

Alluvial Soils

This is the most widely spread and important soil. In fact, the entire northern plains are made of alluvial soil. These have been deposited by three important Himalayan river systems– the Indus, the Ganga and the Brahmaputra. These soils also extend in Rajasthan and Gujarat through a narrow corridor. Alluvial soil is also found in the eastern coastal plains particularly in the deltas of the Mahanadi, the Godavari, the Krishna and the Kaveri rivers.



Fig. 1.6: Alluvial Soil

The alluvial soil consists of various proportions of sand, silt and clay. As we move inlands towards the river valleys, soil particles appear some what bigger in size. In the upper reaches of the river valley i.e. near the place of the break of slope, the soils are coarse. Such soils are more common in piedmont plains such as **Duars**, **Chos** and **Terai**.

Apart from the size of their grains or components, soils are also described on the basis of their age. According to their age alluvial soils can be classified as old alluvial (**Bangar**) and new alluvial (**Khadar**). The **bangar** soil has higher concentration of *kanker* nodules than the **Khadar.** It has more fine particles and is more fertile than the **bangar**.

Alluvial soils as a whole are very fertile. Mostly these soils contain adequate proportion of potash, phosphoric acid and lime which are ideal for the growth of sugarcane, paddy, wheat and other cereal and pulse crops. Due to its high fertility, regions of alluvial soils are intensively cultivated and densely populated. Soils in the drier areas are more alkaline and can be productive after proper treatment and irrigation.

Black Soil

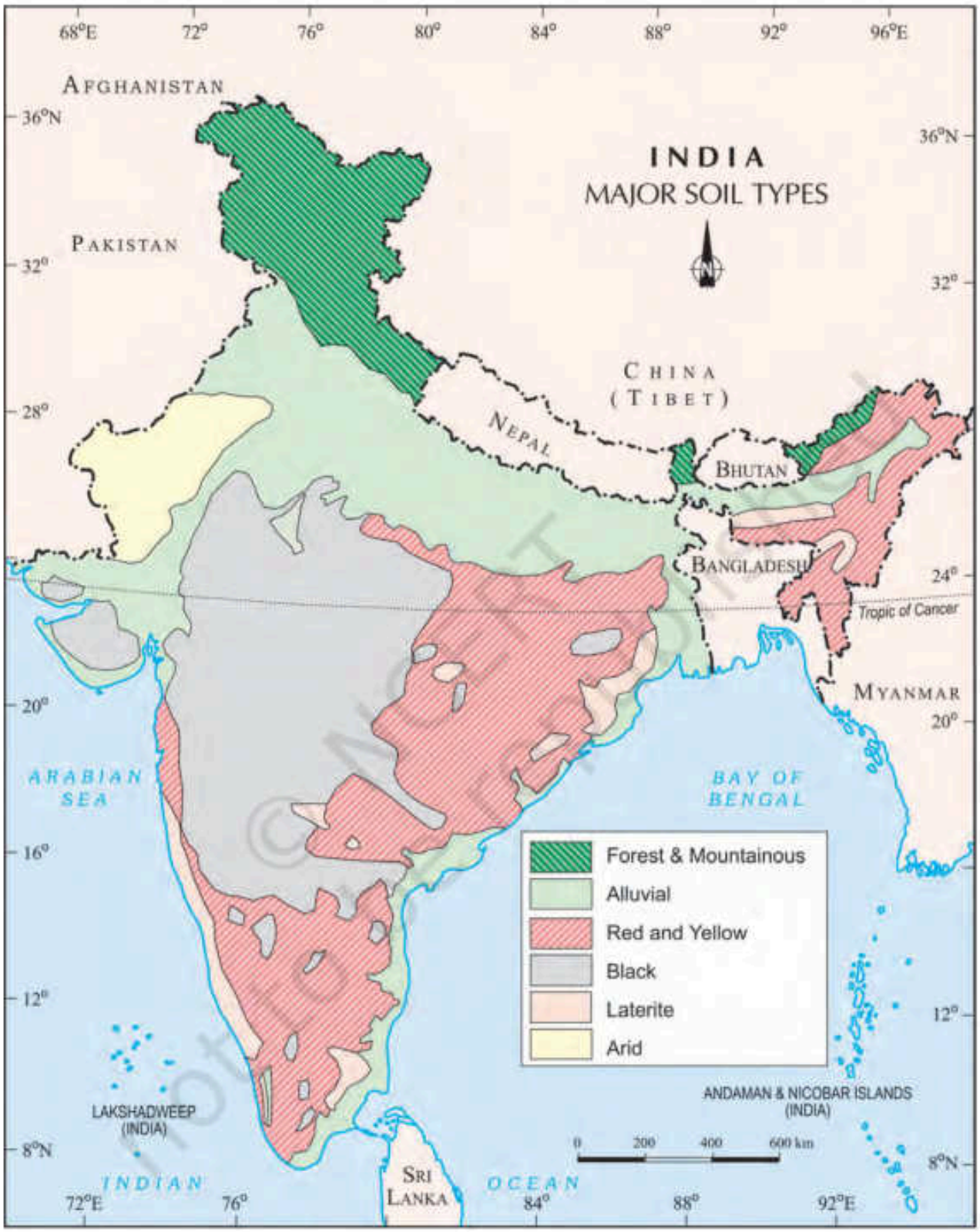
These soils are black in colour and are also known as *regur* soils. Black soil is ideal for growing cotton and is also known as *black cotton soil.* It is believed that climatic condition along with the parent rock material are the important factors for the formation of black soil. This type of soil is typical of the Deccan trap (*Basalt*) region spread over northwest Deccan plateau and is made up of lava flows. They cover the plateaus of Maharashtra, Saurashtra, Malwa, Madhya Pradesh and Chhattisgarh and extend in the south east direction along the Godavari and the Krishna valleys.



Fig. 1.7: Black Soil

The black soils are made up of extremely fine i.e. clayey material. They are well-known for their capacity to hold moisture. In addition, they are rich in soil nutrients, such as calcium carbonate, magnesium, potash and lime. These

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India: Major Soil Types

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soils are generally poor in phosphoric contents. They develop deep cracks during hot weather, which helps in the proper aeration of the soil. These soils are sticky when wet and difficult to work on unless tilled immediately after the first shower or during the pre-monsoon period.

Red and Yellow Soils

Red soil develops on crystalline igneous rocks in areas of low rainfall in the eastern and southern parts of the Deccan plateau. Yellow and red soils are also found in parts of Odisha, Chhattisgarh, southern parts of the middle Ganga plain and along the piedmont zone of the Western Ghats. These soils develop a reddish colour due to diffusion of iron in crystalline and metamorphic rocks. It looks yellow when it occurs in a hydrated form.

Laterite Soil

Laterite has been derived from the Latin word 'later' which means brick. The laterite soil develops under tropical and subtropical climate with alternate wet and dry season. This soil is the result of intense leaching due to heavy rain. Lateritic soils are mostly deep to very deep, acidic (pH<6.0), generally deficient in plant nutrients and occur mostly in southern states, Western Ghats region of Maharashtra, Odisha, some parts of West Bengal and North-east regions. Where these soils support deciduous and evergreen forests, it is humus rich, but under sparse vegetation and in semi-arid environment, it is generally humus poor. They are prone to erosion and degradation due to their position on the landscape. After adopting appropriate soil conservation techniques particularly in the hilly areas of Karnataka, Kerala and Tamil Nadu, this soil is very useful for growing tea and coffee. Red laterite soils in Tamil Nadu, Andhra Pradesh and Kerala are more suitable for crops like cashew nut.



Fig. 1.8: Laterite Soil

Arid Soils

Arid soils range from red to brown in colour. They are generally sandy in texture and saline in nature. In some areas the salt content is very high and common salt is obtained by evaporating the water. Due to the dry climate, high temperature, evaporation is faster and the soil lacks humus and moisture. The lower horizons of the soil are occupied by *Kankar* because of the increasing calcium content downwards. The *Kankar* layer formations in the bottom horizons restrict the infiltration of water. After proper irrigation these soils become cultivable as has been in the case of western Rajasthan.



Fig. 1.9: Arid Soil

Forest Soils

These soils are found in the hilly and mountainous areas where sufficient rain forests are available. The soils texture varies according to the mountain environment where they are formed. They are loamy and silty in valley sides and coarse grained in the upper slopes. In the snow covered areas

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of Himalayas, these soils experience denudation and are acidic with low humus content. The soils found in the lower parts of the valleys particularly on the river terraces and alluvial fans are fertile.

Soil Erosion and Soil Conservation

The denudation of the soil cover and subsequent washing down is described as soil erosion. The processes of soil formation and erosion, go on simultaneously and generally there is a balance between the two. Sometimes, this balance is disturbed due to human activities like deforestation, over-grazing, construction and mining etc., while natural forces like wind, glacier and water lead to soil erosion. The running water cuts through the clayey soils and makes deep channels as **gullies**. The land becomes unfit for cultivation and is known as **bad land**. In the Chambal basin such lands are called ravines. Sometimes water flows as a sheet over large areas down a slope. In such cases the top soil is washed away. This is known as **sheet erosion**. Wind blows loose soil off flat or sloping land known as wind erosion. Soil erosion is also caused due to defective methods of farming. Ploughing in a wrong way i.e. up and down the slope form channels for the quick flow of water leading to soil erosion.



Fig. 1.10: Soil Erosion



Fig. 1.11: Gully Erosion

Ploughing along the contour lines can decelerate the flow of water down the slopes. This is called contour ploughing. Steps can be cut out on the slopes making terraces. Terrace cultivation restricts erosion. Western and central Himalayas have well developed terrace farming. Large fields can be divided into strips. Strips of grass are left to grow between the crops. This breaks up the force of the wind. This method is known as strip cropping. Planting lines of trees to create shelter also works in a similar way. Rows of such trees are called shelter belts. These shelter belts have contributed significantly to the stabilisation of sand dunes and in stabilising the desert in western India.

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Tribal girls using bamboo saplings in a nursery at Mukhali near Silent Valley





Tribal women selling minor forest produce

Leaf litter collection by women folk

Fig. 2.1

Act was implemented in 1972, with various provisions for protecting habitats. An all-India list of protected species was also published. The thrust of the programme was towards protecting the remaining population of certain endangered species by banning hunting, giving legal protection to their habitats, and restricting trade in wildlife. Subsequently, central and many state governments established national parks and wildlife sanctuaries about which you have already studied. The central government also announced several projects for protecting specific animals, which were gravely threatened, including the tiger, the one-horned rhinoceros, the Kashmir stag or *hangul*, three types of crocodiles – fresh water crocodile, saltwater crocodile and the *Gharial*, the Asiatic lion, and others. Most recently, the Indian elephant, black buck (chinkara), the great Indian bustard (*godawan*) and the snow leopard, etc. have been given full or partial legal protection against hunting and trade throughout India.

Project Tiger

Tiger is one of the key wildlife species in the faunal web. In 1973, the authorities realised that the tiger population had dwindled to 1,827 from an estimated 55,000 at the turn of the century. The major threats to tiger population are numerous, such as poaching for trade, shrinking habitat, depletion of prey base species, growing human population, etc. The trade of tiger skins and the use of their bones in traditional medicines, especially in the Asian countries left the tiger population on the verge of extinction. Since India and Nepal provide habitat to about two-thirds of the surviving tiger population in the world, these two nations became prime targets for poaching and illegal trading.

"Project Tiger", one of the well-publicised wildlife campaigns in the world, was launched in 1973. Tiger conservation has been viewed not only as an effort to save an endangered species, but with

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Fig. 2.2: Rhino and deer in Kaziranga National Park

equal importance as a means of preserving biotypes of sizeable magnitude. Corbett National Park in Uttarakhand, Sunderbans National Park in West Bengal, Bandhavgarh National Park in Madhya Pradesh, Sariska Wildlife Sanctuary in Rajasthan, Manas Tiger Reserve in Assam and Periyar Tiger Reserve in Kerala are some of the tiger reserves of India.

The conservation projects are now focusing on biodiversity rather than on a few of its components. There is now a more intensive search for different conservation measures. Increasingly, even insects are beginning to find a place in conservation planning. In the notification under Wildlife Act of 1980 and 1986, several hundred butterflies, moths, beetles, and one dragonfly have been added to the list of protected species. In 1991, for the first time plants were also added to the list, starting with six species.

Activity

Collect more information on the wildlife sanctuaries and national parks of India and cite their locations on the map of India.

Types and Distribution of Forest and Wildlife Resources

Even if we want to conserve our vast forest and wildlife resources, it is rather difficult to manage, control and regulate them. In India, much of its forest and wildlife resources are either owned or managed by the government through the Forest Department or other government departments. These are classified under the following categories.

(i) ***Reserved Forests:*** More than half of the total forest land has been declared **reserved forests**. Reserved forests are regarded as the most valuable as far as the conservation of forest and wildlife resources are concerned.

(ii) ***Protected Forests:*** Almost one-third of the total forest area is protected forest, as declared by the Forest Department. This forest land are protected from any further depletion.

(iii) ***Unclassed Forests:*** These are other forests and wastelands belonging to both government and private individuals and communities.

Reserved and protected forests are also referred to as permanent forest estates maintained for the purpose of producing timber and other forest produce, and for protective reasons. Madhya Pradesh has the largest area under permanent forests, constituting 75 per cent of its total forest area. Jammu and Kashmir, Andhra Pradesh, Uttarakhand, Kerala, Tamil Nadu, West Bengal, and Maharashtra have large percentages of reserved forests of its total forest area whereas Bihar, Haryana, Punjab, Himachal Pradesh, Odisha and Rajasthan have a bulk of it under protected forests. All North-

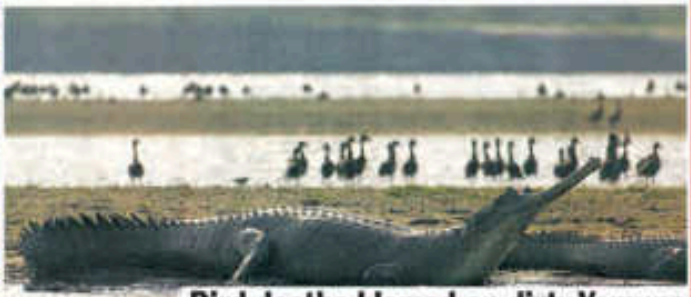
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Gharial on the brink

The gharial population has been at its lowest since the 1970s. What went wrong and what can we do?

ROMULUS WHITAKER
and JANAKI LENIN





Bird deaths blamed on dirty Yamuna

Delhi Govt Report Points To Toxic Elements in Stagnant Water



CRITICALLY ENDANGERED: *Captive gharial at the Madras C*

Can you find out the reasons for the above mentioned problems?

eastern states and parts of Gujarat have a very high percentage of their forests as unclassed forests managed by local communities.

Community and Conservation

Conservation strategies are not new in our country. We often ignore that in India, forests are also home to some of the traditional communities. In some areas of India, local communities are struggling to conserve these habitats along with government officials, recognising that only this will secure their own long-term livelihood. In Sariska Tiger Reserve, Rajasthan, villagers have fought against mining by citing the Wildlife Protection Act. In many areas, villagers themselves are protecting habitats and explicitly rejecting government involvement. The inhabitants of five villages in the Alwar district of Rajasthan have declared 1,200 hectares of forest as the Bhairodev Dakav 'Sonchuri', declaring their own set of rules and regulations which do not allow hunting, and are protecting the wildlife against any outside encroachments.

The famous **Chipko** movement in the Himalayas has not only successfully resisted deforestation in several areas but has also shown that community afforestation with indigenous species can be enormously successful. Attempts to revive the traditional conservation methods or developing new methods of ecological farming are now widespread. Farmers and citizen's groups like the **Beej Bachao Andolan** in Tehri and **Navdanya** have shown that adequate levels of diversified crop production without the use of synthetic chemicals are possible and economically viable.

In India joint forest management (JFM) programme furnishes a good example for involving local communities in the management and restoration of degraded

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Sacred groves - a wealth of diverse and rare species

Nature worship is an age old tribal belief based on the premise that all creations of nature have to be protected. Such beliefs have preserved several virgin forests in pristine form called Sacred Groves (the forests of God and Goddesses). These patches of forest or parts of large forests have been left untouched by the local people and any interference with them is banned.

Certain societies revere a particular tree which they have preserved from time immemorial. The Mundas and the Santhal of Chota Nagpur region worship mahua (**Bassia latifolia**) and kadamba (**Anthocaphalus cadamba**) trees, and the tribals of Odisha and Bihar worship the tamarind (**Tamarindus indica**) and mango (**Mangifera indica**) trees during weddings. To many of us, peepal and banyan trees are considered sacred.

Indian society comprises several cultures, each with its own set of traditional methods of conserving nature and its creations. Sacred qualities are often ascribed to springs, mountain peaks, plants and animals which are closely protected. You will find troops of macaques and langurs around many temples. They are fed daily and treated as a part of temple devotees. In and around Bishnoi villages in Rajasthan, herds of blackbuck, (chinkara), nilgai and peacocks can be seen as an integral part of the community and nobody harms them.

forests. The programme has been in formal existence since 1988 when the state of Odisha passed the first resolution for joint forest management. JFM depends on the formation of local (village) institutions that undertake protection activities mostly on degraded forest land managed by the forest department. In return, the members of these communities are entitled to intermediary benefits like non-timber forest produces and share in the timber harvested by 'successful protection'.

The clear lesson from the dynamics of both environmental destruction and reconstruction in India is that local communities everywhere have to be involved in some kind of natural resource management. But there is still a long way to go before local communities are at the centre-stage in decision-making. Accept only those economic or developmental activities, that are people centric, environment-friendly and economically rewarding.

Activity

Write a short essay on any practices which you may have observed and practised in your everyday lives that conserve and protect the environment around you.

"The tree is a peculiar organism of unlimited kindness and benevolence and makes no demand for its sustenance, and extends generously the products of its life activity. It affords protection to all beings, offering shade even to the axemen who destroy it".

Gautama Buddha (487 B.C.)

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own wells and tube-wells in their farms for irrigation to increase their produce. But have you ever wondered what this could result in? That it may lead to falling groundwater levels, adversely affecting water availability and food security of the people.

Post-independent India witnessed intensive industrialisation and urbanisation, creating vast opportunities for us. Today, large industrial houses are as commonplace as the industrial units of many MNCs (Multinational Corporations). The ever-increasing number of industries has made matters worse by exerting pressure on existing freshwater resources. Industries, apart from being heavy users of water, also require power to run them. Much of this energy comes from hydroelectric power. Moreover, multiplying urban centres with large and dense populations and urban lifestyles have not only added to water and energy requirements but have further aggravated the problem. If you look into the housing societies or colonies in the cities, you would find that most of these have their own groundwater pumping devices to meet their water needs. Not surprisingly, we find that fragile water resources are being over-exploited and have caused their depletion in several of these cities.

Do you know?

Atal Bhujal Yojana (Atal Jal) is being implemented in 8220 water stressed Gram Panchayats of 229 administrative blocks/ talukas in 80 districts of seven states, viz. Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, and Uttar Pradesh. The selected States account for about 37 per cent of the total number of water— stressed (over-exploited, critical and semi-critical) blocks in India. One of the key aspects of Atal Jal is to bring in behavioural changes in the community, from the prevailing attitude of consumption to conservation and smart water management.

Source: *Annual Report, Ministry of Jal Shakti, Government of India 2022–23*

So far we have focused on the quantitative aspects of water scarcity. Now, let us consider another situation where water is sufficiently available to meet the needs of the people, but, the area still suffers from water scarcity. This scarcity may be due to bad quality of water. Lately, there has been a growing concern that even if there is ample water to meet the needs of the people, much of it may be polluted by domestic and industrial wastes, chemicals, pesticides and fertilisers used in agriculture, thus, making it hazardous for human use. Government of India has accorded highest priority to improve the quality of life and enhance ease of living of people especially those living in rual areas by announcing the Jal Jeevan Mission (JJM). The Goal of JJM is to enable every rural household get assured supply of potable piped water at a service level of 55 litres per capita per day regularly on long-term basis by ensuring functionality of the tap water connections. (Source: Economic Survey 2020–21, p.357)

You may have already realised that the need of the hour is to conserve and manage our water resources, to safeguard ourselves from health hazards, to ensure food security, continuation of our livelihoods and productive activities and also to prevent degradation of our natural ecosystems. Over exploitation and mismanagement of water resources will impoverish this resource and cause ecological crisis that may have profound impact on our lives.

Activity

From your everyday experiences, write a short proposal on how you can conserve water.

Multi-purpose River Projects and Integrated Water Resources Management

But, how do we conserve and manage water? Archaeological and historical records show that from ancient times we have been constructing sophisticated hydraulic structures like dams built of stone rubble, reservoirs or lakes, embankments and canals for irrigation. Not surprisingly, we have continued this tradition in modern India by building dams in most of our river basins.

Hydraulic Structures in Ancient India

- In the first century B.C., Sringaverapura near Allahabad had sophisticated water

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harvesting system channelling the flood water of the river Ganga.

- During the time of Chandragupta Maurya, dams, lakes and irrigation systems were extensively built.
- Evidences of sophisticated irrigation works have also been found in Kalinga, (Odisha), Nagarjunakonda (Andhra Pradesh), Bennur (Karnataka), Kolhapur (Maharashtra), etc.
- In the 11th Century, Bhopal Lake, one of the largest artificial lakes of its time was built.
- In the 13th–14th Century, the tank in Hauz Khas, Delhi was constructed by Allauddin Khilji (Khalji) for supplying water to Siri Fort area.

Source: Dying Wisdom, CSE, 1997.



Fig. 3.2: Hirakud Dam

What are dams and how do they help us in conserving and managing water? Dams were traditionally built to impound rivers and rainwater that could be used later to irrigate agricultural fields. Today, dams are built not just for irrigation but for electricity generation, water supply for domestic and industrial uses, flood control, recreation, inland navigation and fish breeding. Hence, dams are now referred to as multi-purpose projects where the many uses of the impounded water are integrated with one another. For example, in the Sutluj-Beas river basin, the Bhakra – Nangal project water is being used both for hydel power production and irrigation. Similarly, the Hirakud project in the Mahanadi basin integrates conservation of water with flood control.

Multi-purpose projects, launched after Independence with their integrated water resources management approach, were thought of as the vehicle that would lead the nation to development and progress, overcoming the handicap of its colonial past. Jawaharlal Nehru proudly proclaimed the dams as the 'temples of modern India'; the reason being that it would integrate development of agriculture and the village economy with rapid industrialisation and growth of the urban economy.

A **dam** is a barrier across flowing water that obstructs, directs or retards the flow, often creating a reservoir, lake or impoundment. "Dam" refers to the reservoir rather than the structure. Most dams have a section called a spillway or weir over which or through which it is intended that water will flow either intermittently or continuously. Dams are classified according to structure, intended purpose or height. Based on structure and the materials used, dams are classified as timber dams, embankment dams or masonry dams, with several subtypes. According to the height, dams can be categorised as large dams and major dams or alternatively as low dams, medium height dams and high dams.

Activity

Find out more about any one traditional method of building dams and irrigation works.

We have sown the crops in Asar
We will bring Bhadu in Bhadra
Floods have swollen the Damodar
The sailing boats cannot sail
Oh! Damodar, we fall at your feet
Reduce the floods a little
Bhadu will come a year later
Let the boats sail on your surface

(This popular Bhadu song in the Damodar valley region narrates the troubles faced by people owing to the flooding of Damodar river known as the river of sorrow.)

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In recent years, multi-purpose projects and large dams have come under great scrutiny and opposition for a variety of reasons. Regulating and damming of rivers affect their natural flow causing poor sediment flow and excessive sedimentation at the bottom of the reservoir, resulting in rockier stream beds and poorer habitats for the rivers' aquatic life. Dams also fragment rivers making it difficult for aquatic fauna to migrate, especially for spawning. The reservoirs that are created on the floodplains also submerge the existing vegetation and soil leading to its decomposition over a period of time.

Do you know?

Sardar Sarovar Dam has been built over the Narmada River in Gujarat. This is one of the largest water resource projects of India covering four states—Maharashtra, Madhya Pradesh, Gujarat and Rajasthan. The Sardar Sarovar project would meet the requirement of water in drought-prone and desert areas. Sardar Sarovar Project will provide irrigation facilities to 18.45 lakh hectare of land, covering 3112 villages in 15 districts of Gujarat. It will also irrigate 2,46,000 hectare of land in the strategic desert districts of Barmer and Jalore in Rajasthan and 37,500 hectare in the tribal hilly tract of Maharashtra through lift. About 75 per cent of the command area in Gujarat is drought prone while entire command in Rajasthan is drought prone. Assured water supply will soon make this area drought proof.

Source: Sardar Sarovar Narmada Nigam Ltd. *https://www.sardarsarovardam.org/*

Ironically, the dams that were constructed to control floods have triggered floods due to sedimentation in the reservoir. Moreover, the big dams have mostly been unsuccessful in controlling floods at the time of excessive rainfall.

Activity

Collect information about floods occurred in different parts of the country due to heavy rainfall in recent times.

These floods have not only devastated life and property but also caused extensive soil erosion. Sedimentation also meant that the flood plains were deprived of silt, a natural fertiliser, further adding on to the problem of land degradation. It was also observed that the multi-purpose projects induced earthquakes, caused water-borne diseases and pests and pollution resulting from excessive use of water.

Irrigation has also changed the cropping pattern of many regions with farmers shifting to water intensive and commercial crops. This has great ecological consequences like salinisation of the soil.

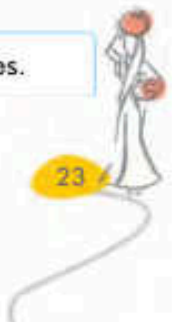
Pradhan Mantri Krishi Sinchayee Yojana has been started which ensures access to some means to protective irrigation for all agricultural farms in the country, thus bringing much desired rural prosperity. Some of the broad objectives of this programme are to enhance the physical access of water on the farm and expand cultivable area under assured irrigation (har khet ko pani), improve on-farm water use efficiency to reduce wastage and increase availability both in duration and extent, irrigation and other water saving technologies (per drop more crop) and introduce sustainable water conservation practices, etc.

Do you know?

Do you know that the Krishna-Godavari dispute is due to the objections raised by Karnataka and Andhra Pradesh governments? It is regarding the diversion of more water at Koyna by the Maharashtra government for a multipurpose project. This would reduce downstream flow in their states with adverse consequences for agriculture and industry.

Activity

Make a list of inter-state water disputes.



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Cropping Pattern

You have studied the physical diversities and plurality of cultures in India. These are also reflected in agricultural practices and cropping patterns in the country. Various types of food and fibre crops, vegetables and fruits, spices and condiments, etc. constitute some of the important crops grown in the country. India has three cropping seasons — **rabi**, **kharif** and **zaid**.

Rabi crops are sown in winter from October to December and harvested in summer from April to June. Some of the important rabi crops are wheat, barley, peas, gram and mustard. Though, these crops are grown in large parts of India, states from the north and north-western parts such as Punjab, Haryana, Himachal Pradesh, Jammu and Kashmir, Uttarakhand and Uttar Pradesh are important for the production of wheat and other rabi crops. Availability of precipitation during winter months due to the western temperate cyclones helps in the success of these crops. However, the success of the green revolution in Punjab, Haryana, western Uttar Pradesh and parts of Rajasthan has also been an important factor in the growth of the above-mentioned rabi crops.

Kharif crops are grown with the onset of monsoon in different parts of the country and these are harvested in September-October. Important crops grown during this season are paddy, maize, jowar, bajra, tur (arhar), moong, urad, cotton, jute, groundnut and soyabean. Some of the most important rice-growing regions are Assam, West Bengal, coastal regions of Odisha, Andhra Pradesh, Telangana, Tamil Nadu, Kerala and Maharashtra, particularly the (Konkan coast) along with Uttar Pradesh and Bihar. Recently, paddy has also become an important crop of Punjab and Haryana. In states like Assam, West Bengal and Odisha, three crops of paddy are grown in a year. These are *Aus*, *Aman* and *Boro*.

In between the rabi and the kharif seasons, there is a short season during the summer months known as the **Zaid** season. Some of the crops produced during 'zaid' are watermelon, muskmelon, cucumber, vegetables and fodder crops. Sugarcane takes almost a year to grow.

Major Crops

A variety of food and non food crops are grown in different parts of the country depending upon the variations in soil, climate and cultivation practices. Major crops grown in India are rice, wheat, millets, pulses, tea, coffee, sugarcane, oil seeds, cotton and jute, etc.

Rice: It is the staple food crop of a majority of the people in India. Our country is the second largest producer of rice in the world after China. It is a kharif crop which requires high temperature, (above 25°C) and high humidity with annual rainfall above 100 cm. In the areas of less rainfall, it grows with the help of irrigation.



Fig. 4.4 (a): Rice Cultivation



Fig. 4.4 (b): Rice is ready to be harvested in the field

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India: Distribution of Rice

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Rice is grown in the plains of north and north-eastern India, coastal areas and the deltaic regions. Development of dense network of canal irrigation and tubewells have made it possible to grow rice in areas of less rainfall such as Punjab, Haryana and western Uttar Pradesh and parts of Rajasthan.

Wheat: This is the second most important cereal crop. It is the main food crop, in north and north-western part of the country. This rabi crop requires a cool growing season and a bright sunshine at the time of ripening. It requires 50 to 75 cm of annual rainfall evenly-distributed over the growing season. There are two important wheat-growing zones in the country – the Ganga-Satluj plains in the north-west and black soil region of the Deccan. The major wheat-producing states are Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Bihar and Rajasthan.



Fig. 4.5: Wheat Cultivation

Millets: Jowar, bajra and ragi are the important millets grown in India. Though, these are known as coarse grains, they have very high nutritional value. For example, ragi is very rich in iron, calcium, other micro nutrients and roughage. Jowar is the third most important food crop with respect to area and production. It is a rain-fed crop mostly grown in the moist areas which hardly needs irrigation. Major Jowar producing States are Maharashtra, Karnataka, Andhra Pradesh and Madhya Pradesh.



Fig. 4.6: Bajra Cultivation

Bajra grows well on sandy soils and shallow black soil. Major Bajra producing States are Rajasthan, Uttar Pradesh, Maharashtra, Gujarat and Haryana. Ragi is a crop of dry regions and grows well on red, black, sandy, loamy and shallow black soils. Major ragi producing states are: Karnataka, Tamil Nadu, Himachal Pradesh, Uttarakhand, Sikkim, Jharkhand and Arunachal Pradesh.

Maize: It is a crop which is used both as food and fodder. It is a kharif crop which requires temperature between 21°C to 27°C and grows well in old alluvial soil. In some states like Bihar maize is grown in rabi season also. Use of modern inputs such as HYV seeds, fertilisers and irrigation have contributed to the increasing production of maize. Major maize-producing states are Karnataka, Madhya Pradesh, Uttar Pradesh, Bihar, Andhra Pradesh and Telangana.



Fig. 4.7: Maize Cultivation

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India is an important producer of pea, cauliflower, onion, cabbage, tomato, brinjal and potato.

Non-Food Crops

Rubber: It is an equatorial crop, but under special conditions, it is also grown in tropical and sub-tropical areas. It requires moist and humid climate with rainfall of more than 200 cm. and temperature above 25°C.

Rubber is an important industrial raw material. It is mainly grown in Kerala, Tamil Nadu, Karnataka and Andaman and Nicobar islands and Garo hills of Meghalaya.

Activity

List the items which are made of rubber and are used by us.

Fibre Crops: Cotton, jute, hemp and natural silk are the four major fibre crops grown in India. The first three are derived from the crops grown in the soil, the latter is obtained from cocoons of the silkworms fed on green leaves specially mulberry. Rearing of silk worms for the production of silk fibre is known as sericulture.

Cotton: India is believed to be the original home of the cotton plant. Cotton is one of the main raw materials for cotton textile industry. India is second largest producer of cotton after China. Cotton grows well in drier parts of the black cotton soil of the Deccan plateau. It requires high temperature, light rainfall or irrigation, 210 frost-free days and bright sun-shine for its growth. It is a kharif crop and requires 6 to 8 months to mature. Major cotton-producing states are– Maharashtra, Gujarat, Madhya Pradesh, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Punjab, Haryana and Uttar Pradesh.



Fig. 4.14: Cotton Cultivation

Jute: It is known as the golden fibre. Jute grows well on well-drained fertile soils in the flood plains where soils are renewed every year. High temperature is required during the time of growth. West Bengal, Bihar, Assam, Odisha and Meghalaya are the major jute producing states. It is used in making gunny bags, mats, ropes, yarn, carpets and other artefacts.

Technological and Institutional Reforms

It was mentioned in the previous pages that agriculture has been practised in India for thousands of years. Sustained uses of land without compatible techno-institutional changes have hindered the pace of agricultural development. Inspite of development of sources of irrigation most of the farmers in large parts of the country still depend upon monsoon and natural fertility in order to carry on their agriculture. For a growing population, this poses a serious challenge. Agriculture which provides livelihood for more than 60 per cent of its population, needs some serious technical and institutional reforms. Thus, collectivisation, consolidation of holdings, cooperation and abolition of zamindari, etc. were given priority to bring about institutional reforms in the country after Independence. 'Land reform' was the main focus of our First Five Year Plan. The right of inheritance had already led to fragmentation of land holdings necessitating consolidation of holdings.

The laws of land reforms were enacted but the implementation was lacking or lukewarm. The Government of India embarked upon introducing agricultural reforms to improve Indian agriculture in the 1960s and 1970s. The Green Revolution based on the use of package technology and the White Revolution (Operation Flood) were some of the strategies initiated to improve the lot of Indian agriculture. But, this too led to the concentration of development in few selected areas. Therefore, in the 1980s and 1990s, a comprehensive land development programme was initiated, which included both institutional and technical

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Fig. 4.15: Modern technological equipments used in agriculture

reforms. Provision for crop insurance against drought, flood, cyclone, fire and disease, establishment of Grameen banks, cooperative societies and banks for providing loan facilities to the farmers at lower rates of interest were some important steps in this direction.

Kisan Credit Card (KCC), Personal Accident Insurance Scheme (PAIS) are some other schemes introduced by the Government of India for the benefit of the farmers. Moreover, special weather bulletins and agricultural programmes for farmers were introduced on the radio and television. The government also announces minimum support price, remunerative and procurement prices for important crops to check the exploitation of farmers by speculators and middlemen.

Activity

Collect information about agriculture, horticulture, agricultural schemes, etc. from *Farmers' Portal* website https://farmer.gov.in/FarmerHome.aspx. Discuss about the benefits of the information available on the portal.

Bhoodan – Gramdan

Mahatma Gandhi declared Vinoba Bhave as his spiritual heir. He also participated in Satyagraha as one of the foremost satyagrahis. He was one of the votaries of Gandhi's concept of **gram swarajya**. After Gandhiji's martyrdom, Vinoba Bhave undertook **padyatra** to spread Gandhiji's message covered almost the entire country. Once, when he was delivering a lecture at Pochampally in Telangana, some poor landless villagers demanded some land for their economic well-being. Vinoba Bhave could not promise it to them immediately but assured them to talk to the Government of India regarding provision of land for them if they undertook cooperative farming.

Suddenly, Shri Ram Chandra Reddy stood up and offered 80 acres of land to be distributed among 80 land-less villagers. This act was known as 'Bhoodan'. Later he travelled and introduced his ideas widely all over India. Some zamindars, owners of many villages offered to distribute some villages among the landless. It was known as **Gramdan**. However, many land-owners chose to provide some part of their land to the poor farmers due to the fear of land ceiling act. This Bhoodan-Gramdan movement initiated by Vinoba Bhave is also known as the **Blood-less Revolution**.

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You have already learnt about rocks. Rocks are combinations of homogenous substances called **minerals**. Some rocks, for instance limestone, consist of a single mineral only, but majority of the rock consist of several minerals in varying proportions. Although, over 2000 minerals have been identified, only a few are abundantly found in most of the rocks.

A particular mineral that will be formed from a certain combination of elements depends upon the physical and chemical conditions under which the material forms. This, in turn, results in a wide range of colours, hardness, crystal forms, lustre and density that a particular mineral possesses. Geologists use these properties to classify the minerals.

Study of Minerals by Geographers and Geologists

Geographers study minerals as part of the earth's crust for a better understanding of landforms. The distribution of mineral resources and associated economic activities are of interest to geographers. A geologist, however, is interested in the formation of minerals, their age and physical and chemical composition.

However, for general and commercial purposes minerals can be classified as under.

Mode of Occurrence of Minerals

Where are these minerals found?

Minerals are usually found in "ores". The term ore is used to describe an accumulation of any mineral mixed with other elements. The mineral content of the ore must be in sufficient concentration to make its extraction commercially viable. The type of formation or structure in which they are found determines the relative ease with which mineral ores may be mined. This also determines the cost of extraction. It is, therefore, important for us to understand the main types of formations in which minerals occur.

Minerals generally occur in these forms:

(i) In igneous and metamorphic rocks minerals may occur in the cracks, crevices, faults or joints. The smaller occurrences are called **veins** and the larger are called **lodes**. In most cases, they are formed when minerals in liquid/molten and gaseous forms are forced upward through cavities towards the earth's surface. They cool and solidify as they rise. Major metallic minerals like tin, copper, zinc and lead etc. are obtained from veins and lodes.

(ii) In sedimentary rocks a number of minerals occur in **beds** or **layers**. They have been formed as a result of deposition, accumulation and concentration in horizontal strata. Coal and some forms of iron ore have been concentrated as a result of long periods under great heat and pressure. Another group of sedimentary minerals include gypsum, potash salt and sodium salt. These are formed as a result of **evaporation** especially in arid regions.

(iii) Another mode of formation involves the decomposition of surface rocks, and the removal of soluble constituents, leaving a **residual mass of weathered material** containing ores. Bauxite is formed this way.

Classification of Minerals

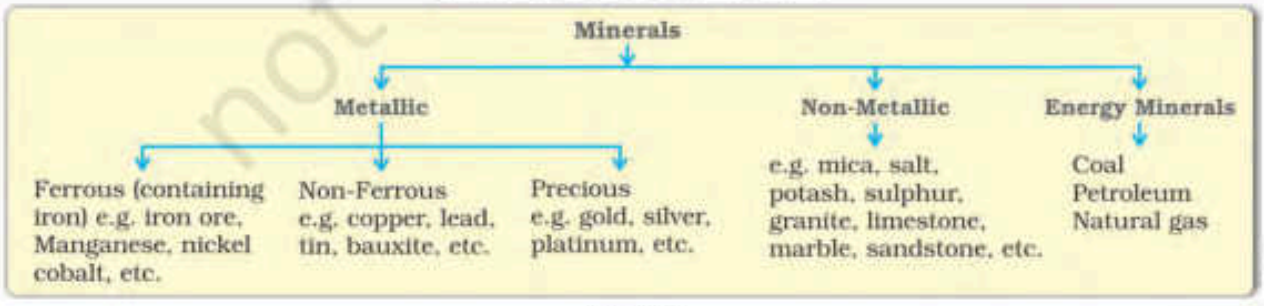


Fig. 5.1

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(iv) Certain minerals may occur as **alluvial deposits** in sands of valley floors and the base of hills. These deposits are called 'placer deposits' and generally contain minerals, which are not corroded by water. Gold, silver, tin and platinum are most important among such minerals.

(v) The ocean waters contain vast quantities of minerals, but most of these are too widely diffused to be of economic significance. However, common salt, magnesium and bromine are largely derived from ocean waters. The ocean beds, too, are rich in manganese nodules.

Interesting Fact

Rat-Hole Mining. Do you know that most of the minerals in India are nationalised and their extraction is possible only after obtaining due permission from the government? But in most of the tribal areas of the north-east India, minerals are owned by individuals or communities. In Meghalaya, there are large deposits of coal, iron ore, limestone and dolomite etc. Coal mining in Jowai and Cherapunjee is done by family member in the form of a long narrow tunnel, known as 'Rat hole' mining. The National Green Tribunal has declared such activities illegal and recommended that these should be stopped forthwith.

Dig a little deeper: What is the difference between an open pit mine, a quarry and an underground mine with shafts?

India is fortunate to have fairly rich and varied mineral resources. However, these are unevenly distributed. Broadly speaking, peninsular rocks contain most of the reserves of coal, metallic minerals, mica and many other non-metallic minerals. Sedimentary rocks on the western and eastern flanks of the peninsula, in Gujarat and Assam have most of the petroleum deposits. Rajasthan with the rock systems of the peninsula, has reserves of many non-ferrous minerals. The vast alluvial plains of north India are almost devoid of economic minerals. These variations exist largely because of the differences in the geological structure, processes and time involved in the formation of minerals.

Let us now study the distribution of a few major minerals in India. Always remember that the concentration of mineral in the ore, the ease of extraction and closeness to the market play an important role in affecting the economic viability of a reserve. Thus, to meet the demand, a choice has to be made between a number of possible options. When this is done a mineral 'deposit' or 'reserve' turns into a **mine**.

Ferrous Minerals

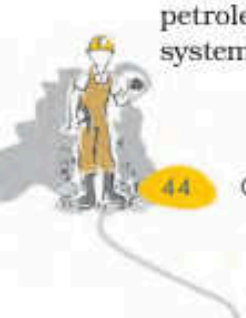
Ferrous minerals account for about three-fourths of the total value of the production of metallic minerals. They provide a strong base for the development of metallurgical industries. India exports substantial quantities of ferrous minerals after meeting her internal demands.

Iron Ore

Iron ore is the basic mineral and the backbone of industrial development. India is endowed with fairly abundant resources of iron ore. India is rich in good quality iron ores. Magnetite is the finest iron ore with a very high content of iron up to 70 per cent. It has excellent magnetic qualities, especially valuable in the electrical industry. Hematite ore is the most important industrial iron ore in terms of the quantity used, but has a slightly lower iron content than magnetite. (50-60 per cent). In 2018–19 almost entire production of iron ore (97%) accrued from Odisha, Chhattisgarh, Karnataka and Jharkhand. The remaining production (3%) was from other states.

Do you know?

Kudre in Kannada means horse. The highest peak in the western ghats of Karnataka resembles the face of a horse. The Bailadila hills look like the hump of an ox, and hence its name.



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Fig. 5.2: Iron ore mine

The major iron ore belts in India are:

- *Odisha-Jharkhand belt*: In Odisha high grade hematite ore is found in Badampahar mines in the Mayurbhanj and Kendujhar districts. In the adjoining Singbhum district of Jharkhand haematite iron ore is mined in Gua and Noamundi.
- *Durg-Bastar-Chandrapur belt* lies in Chhattisgarh and Maharashtra. Very high grade hematites are found in the famous Bailadila range of hills in the Bastar district of Chhattisgarh. The range of hills comprise of 14 deposits of super high grade hematite iron ore. It has the best physical properties needed for steel making. Iron ore from these mines is exported to Japan and South Korea via Vishakhapatnam port.
- *Ballari-Chitradurga-Chikkamagaluru-Tumakuru belt* in Karnataka has large reserves of iron ore. The Kudremukh mines located in the Western Ghats of Karnataka are a 100 per cent export unit. Kudremukh deposits are known to be one of the largest in the world. The ore is transported as slurry through a pipeline to a port near Mangaluru.
- *Maharashtra-Goa belt* includes the state of Goa and Ratnagiri district of Maharashtra. Though, the ores are not of very high quality, yet they are efficiently exploited. Iron ore is exported through Marmagao port.

Manganese

Manganese is mainly used in the manufacturing of steel and ferro-manganese alloy. Nearly 10 kg of manganese is required to manufacture one tonne of steel. It is also used in manufacturing bleaching powder, insecticides and paints.

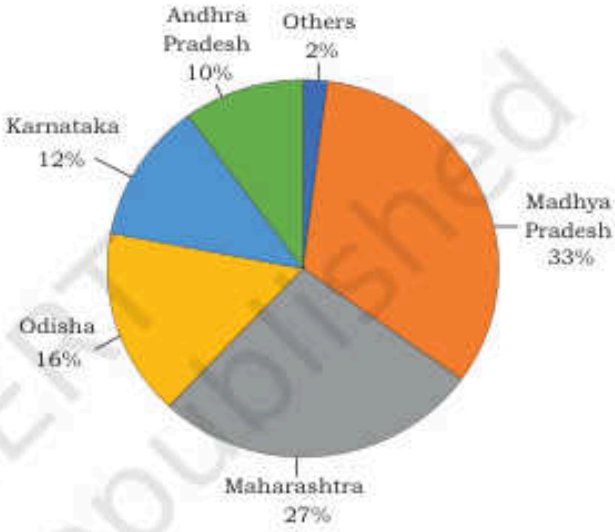


Fig. 5.3: Production of Manganese showing state-wise share in per cent, 2018–19

Dig a little deeper: Superimpose the maps showing distribution of iron ore, manganese, coal and iron and steel industry. Do you see any correlation. Why?

Non-Ferrous Minerals

India's reserves and production of non-ferrous minerals is not very satisfactory. However, these minerals, which include copper, bauxite, lead, zinc and gold play a vital role in a number of metallurgical, engineering and electrical industries. Let us study the distribution of copper and bauxite.

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India: Distribution of Iron Ore, Manganese, Bauxite and Mica

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Copper

India is critically deficient in the reserve and production of copper. Being malleable, ductile and a good conductor, copper is mainly used in electrical cables, electronics and chemical industries. The Balaghat mines in Madhya Pradesh, Khetri mines in Rajasthan and Singhbhum district of Jharkhand are leading producers of copper.



Fig. 5.4: Copper mines at Malanjkhand

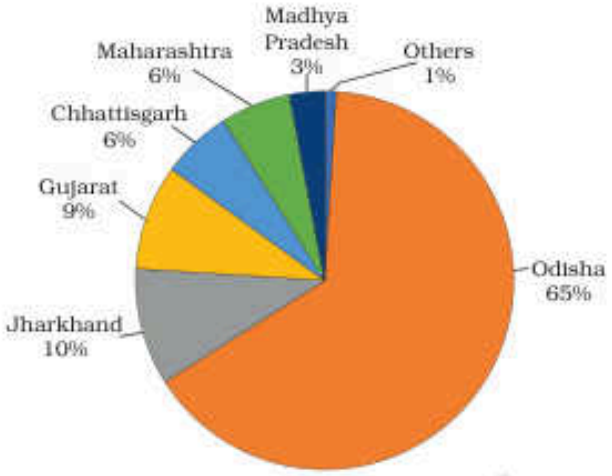


Fig. 5.5: Production of Bauxite showing state-wise share in per cent, 2018–19

Bauxite

Though, several ores contain aluminium, it is from bauxite, a clay-like substance that alumina and later aluminium is obtained. Bauxite deposits are formed by the decomposition of a wide variety of rocks rich in aluminium silicates.

Aluminium is an important metal because it combines the strength of metals such as iron, with extreme lightness and also with good conductivity and great malleability.

India's bauxite deposits are mainly found in the Amarkantak plateau, Maikal hills and the plateau region of Bilaspur-Katni.

Odisha was the largest bauxite producing state in India in 2018-19. Panchpatmali deposits in Koraput district are the most important bauxite deposits in the state.



Fig.5.6: Bauxite Mine

Dig a little deeper: Locate the mines of Bauxite on the physical map of India.



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Interesting Fact

After the discovery of aluminium Emperor Napoleon III wore buttons and hooks on his clothes made of aluminium and served food to his more illustrious guests in aluminium utensils and the less honourable ones were served in gold and silver utensils. Thirty years after this incident aluminium bowls were most common with the beggars in Paris.

Rock Minerals

Limestone is found in association with rocks composed of calcium carbonates or calcium and magnesium carbonates. It is found in sedimentary rocks of most geological formations. Limestone is the basic raw material for the cement industry and essential for smelting iron ore in the blast furnace.

Dig a little deeper: Study the maps to explain why Chota Nagpur is a storehouse of minerals.

Non-Metallic Minerals

Mica is a mineral made up of a series of plates or leaves. It splits easily into thin sheets. These sheets can be so thin that a thousand can be layered into a mica sheet of a few centimeters high. Mica can be clear, black, green, red yellow or brown. Due to its excellent di-electric strength, low power loss factor, insulating properties and resistance to high voltage, mica is one of the most indispensable minerals used in electric and electronic industries.

Mica deposits are found in the northern edge of the Chota Nagpur plateau. Koderma Gaya – Hazaribagh belt of Jharkhand is the leading producer.

In Rajasthan, the major mica producing area is around Ajmer. Nellore mica belt of Andhra Pradesh is also an important producer in the country.

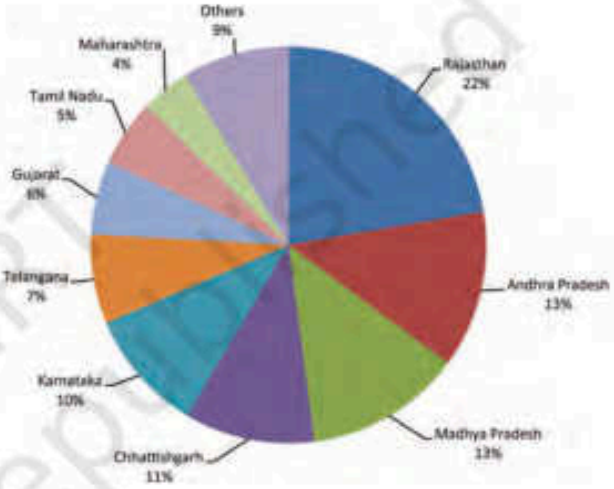


Fig. 5.7: Production of Limestone showing state-wise share in per cent, 2018–19

Hazards of Mining

Have you ever wondered about the efforts the miners make in making life comfortable for you? What are the impacts of mining on the health of the miners and the environment?

The dust and noxious fumes inhaled by miners make them vulnerable to pulmonary diseases. The risk of collapsing mine roofs, inundation and fires in coalmines are a constant threat to miners.

The water sources in the region get contaminated due to mining. Dumping of waste and slurry leads to degradation of land, soil, and increase in stream and river pollution.

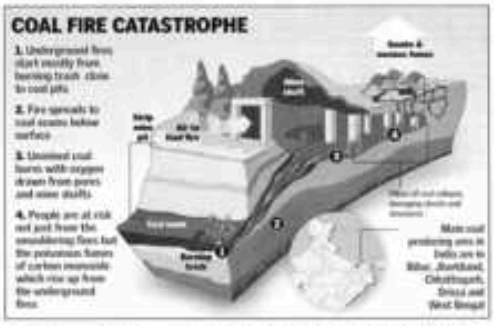


Fig. 5.8: Air pollution due to generation of dust in mining areas



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Jharia to be shifted



Law Kumar Mishra | TNN

Dhanbad: The government has decided to relocate an entire township — Jharia — because of the uncontrollable spread of underground fires that have been burning for years in the coal belt.

"The Centre has already sanctioned Rs 14 crore for the first phase of shifting."

Claiming it to be one of the world's major evacuation drives, Soren said the shifting would be done in three phases up to Free score and he also announced that the rest of shifting people will be at Rs 20,000 crore. The Rana panel had also said, "It is profitable to shift people from Jharia, since coal worth Rs 60,000 crore is lying unutilised beneath the Jharia

Coal unit fined Rs 300 cr in damages

Dhananjay Mahapatra | TNN

New Delhi: The Supreme Court has asked India's largest coal producing company — South Eastern Coalfields Ltd — to pay within a month Rs 300 crore as compensation for using forest land in Chhattisgarh. This is in addition to Rs 50 crore already paid by the PSU, which is a subsidiary of Coal India Ltd, to continue its operations in the state.

SECL will have to pay Rs 100 crore within a week and the rest within a month, the Green Bench comprising CJI Y K Sabharwal and Justices Arijit Pasayat and S H Kapadia directed on Friday.

The court, in its earlier judgment, had said that all companies were liable to pay Penal Compensatory Afforestation cost and amount of Net Present Value to continue operations in forest land already allotted to them by the states.

Appearing for the PSU, solicitor general G E Vahanvati argued that the company had one of the best records for protection of environment and had won awards at the national and state level. He said the Rs 50 crore already deposited by the company was compensation enough for it to be allowed coal mining operations. Appearing for the ministry of environment and forests, counsel A D N Rao

'Over 50% of coal-belt mines unsafe'

Law Kumar Mishra | TNN

Dhanbad: Three days after one of the worst mining disasters in decades left 54 people dead, the chief of the company which owns these mines, said no less than half the mines in the area did not meet the basic safety standards.

Bharat Coking Coal Ltd (BCCL) chairman Partho S Bhattacharya, however, said on Saturday the company shouldn't be blamed for operating unsafe mines because workers and trade unions had blocked moves to shut these for fear of losing jobs. He said only one out of 41 mines was operated scientifically and of the rest, the many were functioning against the advice of the Directorate General of Mines Safety. He described safety standards in 24 mines as poor.

Bhattacharya said workers were aware that they would have to seek voluntary retirement if unsafe mines were shut.

Bharat Coking Coal management on Saturday also handed over employment letters to the immediate people nominated by families of victims. Ironically, these people, most likely, would go to work in mines deemed unsafe by the company.

BCCL's audit of the safety status of mines, graded second and third degrees,



WAIT AND WATCH: A crowd watches the ongoing rescue operation outside the Bharat Coking Coal Ltd mine at Bhatdih in Jharkhand on Thursday

'Safety standards were not in place'

Law Kumar Mishra | TNN

Dhanbad: Union coal minister Shibu Soren on Thursday announced an ex gratia of Rs 3 lakh to the families of the mine tragedy victims. Soren said: "Dependents of the victims will be provided employment by the Bharat Coking Coal from today. A probe by the director general of mines safety and the labour commissioner will also be initiated."

Soren also held discussions with Bharat Coking Coal officials and promised medical help to the four survivors.

tained, even though the director general of mine safety carries out surveys of the safety norms," an expert said.

Some agitating miners alleged that "there is no emergency measures inside the mine. We go deep — between 400 feet and 1,500 ft — into the mines with only an oxygen mask." The tragedy points to the lack of security measures by Bharat Coking Coal despite previous instances of disasters due to methane leak.

At least 48 people were killed in explosions in Jeetpur mine, 43 in Sudamdeeh and 30 in

Stricter safety regulations and implementation of environmental laws are essential to prevent mining from becoming a "killer industry".

Conservation of Minerals

We all appreciate the strong dependence of industry and agriculture upon mineral deposits and the substances manufactured from them. The total volume of workable mineral deposits is an insignificant fraction i.e. one per cent of the earth's crust. We are rapidly consuming mineral resources that required millions of years to be created and concentrated. The geological processes of mineral formation are so slow that the rates of replenishment are infinitely small in comparison to the present rates of consumption. Mineral resources are, therefore, finite and non-renewable. Rich mineral deposits are our country's extremely valuable but short-lived possessions. Continued extraction of ores leads to increasing costs as mineral extraction comes from greater depths along with decrease in quality.

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A concerted effort has to be made in order to use our mineral resources in a planned and sustainable manner. Improved technologies need to be constantly evolved to allow use of low grade ores at low costs. Recycling of metals, using scrap metals and other substitutes are steps in conserving our mineral resources for the future.

Dig a little deeper: Make a list of items where substitutes are being used instead of minerals. Where are these substitutes obtained from?

Energy Resources

Energy is required for all activities. It is needed to cook, to provide light and heat, to propel vehicles and to drive machinery in industries.

Energy can be generated from fuel minerals like coal, petroleum, natural gas, uranium and from electricity. Energy resources can be classified as conventional and non-conventional sources. Conventional sources include: firewood, cattle dung cake, coal, petroleum, natural gas and electricity (both hydel and thermal). Non-conventional sources include solar, wind, tidal, geothermal, biogas and atomic energy. Firewood and cattle dung cake are most common in rural India. According to one estimate more than 70 per cent energy requirement in rural households is met by these two ; continuation of these is increasingly becoming difficult due to decreasing forest area. Moreover, using dung cake too is being discouraged because it consumes most valuable manure which could be used in agriculture.

Conventional Sources of Energy

Coal: In India, coal is the most abundantly available fossil fuel. It provides a substantial part of the nation's energy needs. It is used for power generation, to supply energy to industry as well as for domestic needs. India is highly dependent on coal for meeting its commercial energy requirements.

As you are already aware that coal is formed due the compression of plant material over millions of years. Coal, therefore, is found in a variety of forms depending on the degrees of compression and the depth and time of burial. Decaying plants in swamps produce peat. Which has a low carbon and high moisture contents and low heating capacity. **Lignite** is a low grade brown coal, which is soft with high moisture content. The principal lignite reserves are in Neyveli in Tamil Nadu and are used for generation of electricity. Coal that has been buried deep and subjected to increased temperatures is **bituminous** coal. It is the most popular coal in commercial use. Metallurgical coal is high grade bituminous coal which has a special value for smelting iron in blast furnaces. **Anthracite** is the highest quality hard coal.

In India coal occurs in rock series of two main geological ages, namely Gondwana, a little over 200 million years in age and in tertiary deposits which are only about 55 million years old. The major resources of Gondwana coal, which are metallurgical coal, are located in Damodar valley (West Bengal-



Fig. 5.9 (a): A view from inside of a coal mine



Fig. 5.9 (b): A view from outside of a coal mine

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India: Distribution of Coal, Oil and Natural Gas

Activity

Collect information about cross country natural gas pipelines laid by GAIL (India) under "One Nation One Grid".

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Jharkhand). Jharia, Raniganj, Bokaro are important coalfields. The Godavari, Mahanadi, Son and Wardha valleys also contain coal deposits.

Tertiary coals occur in the north eastern states of Meghalaya, Assam, Arunachal Pradesh and Nagaland.

Remember coal is a bulky material, which loses weight on use as it is reduced to ash. Hence, heavy industries and thermal power stations are located on or near the coalfields.

Petroleum

Petroleum or mineral oil is the next major energy source in India after coal. It provides fuel for heat and lighting, lubricants for machinery and raw materials for a number of manufacturing industries. Petroleum refineries act as a "nodal industry" for synthetic textile, fertiliser and numerous chemical industries.

Most of the petroleum occurrences in India are associated with anticlines and fault traps in the rock formations of the tertiary age. In regions of folding, anticlines or domes, it occurs where oil is trapped in the crest of the upfold. The oil bearing layer is a porous limestone or sandstone through which oil may flow. The oil is prevented from rising or sinking by intervening non-porous layers.

Petroleum is also found in fault traps between porous and non-porous rocks. Gas, being lighter usually occurs above the oil.

Mumbai High, Gujarat and Assam are major petroleum production areas in India. From the map locate the 3 major off shore fields of western India. Ankeleshwar is the most important field of Gujarat. Assam is the oldest oil producing state of India. Digboi, Naharkatiya and Moran-Hugrijan are the important oil fields in the state.

Natural Gas

Natural Gas is found with petroleum deposits and is released when crude oil is brought to the surface. It can be used as a domestic and industrial fuel. It is used as fuel in power sector to generate electricity, for heating purpose in industries, as raw material in chemical, petrochemical and fertilizer industries, as transport fuel and as cooking fuel. With the expansion of gas infrastructure and local city gas distribution (COD) networks, natural gas is also emerging as a preferred transport fuel (CNG) and cooking fuel (PNG) at homes. India's major gas reserves are found in the Mumbai High and allied fields along the west coast which are supplemented by finds in the Cambay basin. Along the East Coast, new reserves of natural gas have been discovered in the Krishna-Godavari basin.

The first 1,700 km long Hazira-Vijaipur-Jagdishpur (HVJ) cross country gas pipeline, constructed by GAIL (India), linked Mumbai High and Bassein gas fields with various fertilizer, power and industrial complexes in western and northen India. This artery provided impetus to Indian gas market development. Overall, India's gas infrastructure has expanded over ten times from 1,700 km to 18,500 km of cross-country pipelines and is expected to soon reach over 34, 000 km as Gas Grid by linking all gas sources and consuming markets across the country including North Eastern states.

Electricity

Electricity has such a wide range of applications in today's world that, its percapita consumption is considered as an index of development. Electricity is generated mainly in two ways: by running water which drives hydro turbines to generate *hydro electricity;* and by burning other fuels such as coal, petroleum and natural gas to drive turbines to produce *thermal power.* Once generated the electricity is exactly the same.

Activity

Name some river valley projects and write the names of the dams built on these rivers.

Hydro electricity is generated by fast flowing water, which is a renewable resource. India has a number of multi-purpose projects like the Bhakra Nangal, Damodar Valley corporation, the Kopili Hydel Project etc. producing hydroelectric power.

Thermal electricity is generated by using coal, petroleum and natural gas. The thermal power stations use non-renewable fossil fuels for generating electricity.



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manufacturing industry has not only assisted agriculturists in increasing their production but also made the production processes very efficient.

In the present day world of globalisation, our industry needs to be more efficient and competitive. Self-sufficiency alone is not enough. Our manufactured goods must be at par in quality with those in the international market. Only then, will we be able to compete in the international market.

Classification of Industries

List the various manufactured products you use in your daily life such as – transistors, electric bulbs, vegetable oil, cement, glassware, petrol, matches, scooters, automobiles, medicines and so on. If we classify the various industries based on a particular criterion then we would be able to understand their manufacturing better. Industries may be classified as follows:

On the basis of source of raw materials used:

- Agro based: cotton, woollen, jute, silk textile, rubber and sugar, tea, coffee, edible oil.
- Mineral based: iron and steel, cement, aluminium, machine tools, petrochemicals.

According to their main role:

- Basic or key industries are those which supply their products as raw materials to manufacture other goods e.g. iron and steel and copper smelting, aluminum smelting.
- Consumer industries that produce goods for direct use by consumers – sugar, toothpaste, paper, sewing machines, fans etc.

On the basis of capital investment:

- A small scale industry is defined with reference to the maximum investment allowed on the assets of a unit. This limit has changed over a period of time. At present the maximum investment allowed is rupees one crore.

On the basis of ownership:

- Public sector, owned and operated by government agencies – BHEL, SAIL etc.
- Private sector industries owned and operated by individuals or a group of individuals –TISCO, Bajaj Auto Ltd., Dabur Industries.
- Joint sector industries which are jointly run by the state and individuals or a group of individuals. Oil India Ltd. (OIL) is jointly owned by public and private sector.
- Cooperative sector industries are owned and operated by the producers or suppliers of raw materials, workers or both. They pool in the resources and share the profits or losses proportionately. Such examples are the sugar industry in Maharashtra, the coir industry in Kerala.

Based on the bulk and weight of raw material and finished goods:

- Heavy industries such as iron and steel
- Light industries that use light raw materials and produce light goods such as electrical goods industries.

Activity

Classify the following into two groups on the basis of bulk and weight of raw material and finished goods.

(i) Oil	(vi) Sewing Machines
(ii) Knitting needles	(vii) Shipbuilding
(iii) Brassware	(viii) Electric Bulbs
(iv) Fuse wires	(ix) Paint brushes
(v) Watches	(x) Automobiles

Agro-based Industries

Cotton, jute, silk, woollen textiles, sugar and edible oil, etc. industries are based on agricultural raw materials.



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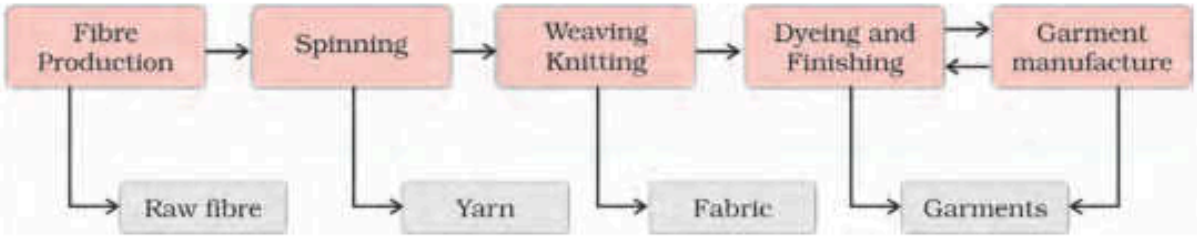


Fig. 6.1: Value addition in the textile industry

Textile Industry: The textile industry occupies unique position in the Indian economy, because it contributes significantly to industrial production, employment generation and foreign exchange earnings. It is the only industry in the country, which is self-reliant and complete in the value chain i.e., from raw material to the highest value added products.

Cotton Textiles: In ancient India, cotton textiles were produced with hand spinning and handloom weaving techniques. After the 18th century, power-looms came into use. Our traditional industries suffered a setback during the colonial period because they could not compete with the mill-made cloth from England.

- The first successful textile mill was established in Mumbai in 1854.
- The two world wars were fought in Europe, India was a British colony. There was a demand for cloth in U.K. hence, they gave a boost to the development of the cotton textile industry.

In the early years, the cotton textile industry was concentrated in the cotton growing belt of Maharashtra and Gujarat. Availability of raw cotton, market, transport including accessible port facilities, labour, moist climate, etc. contributed towards its localisation. This industry has close links with agriculture and provides a living to farmers, cotton boll pluckers and workers engaged in ginning, spinning, weaving, dyeing, designing, packaging, tailoring and sewing. The industry by creating demands supports many other industries, such as, chemicals and dyes, packaging materials and engineering works.

While spinning continues to be centralised in Maharashtra, Gujarat and Tamil Nadu, weaving is highly decentralised to provide scope for incorporating traditional skills and designs of weaving in cotton, silk, zari, embroidery, etc. India has world class production in spinning, but weaving supplies low quality of fabric as it cannot use much of the high quality yarn produced in the country. Weaving is done by handloom, powerloom and in mills.

The handspun khadi provides large scale employment to weavers in their homes as a cottage industry.

Why did Mahatma Gandhi lay emphasis on spinning yarn and weaving khadi?

Why is it important for our country to keep the mill sector loomage lower than power loom and handloom?

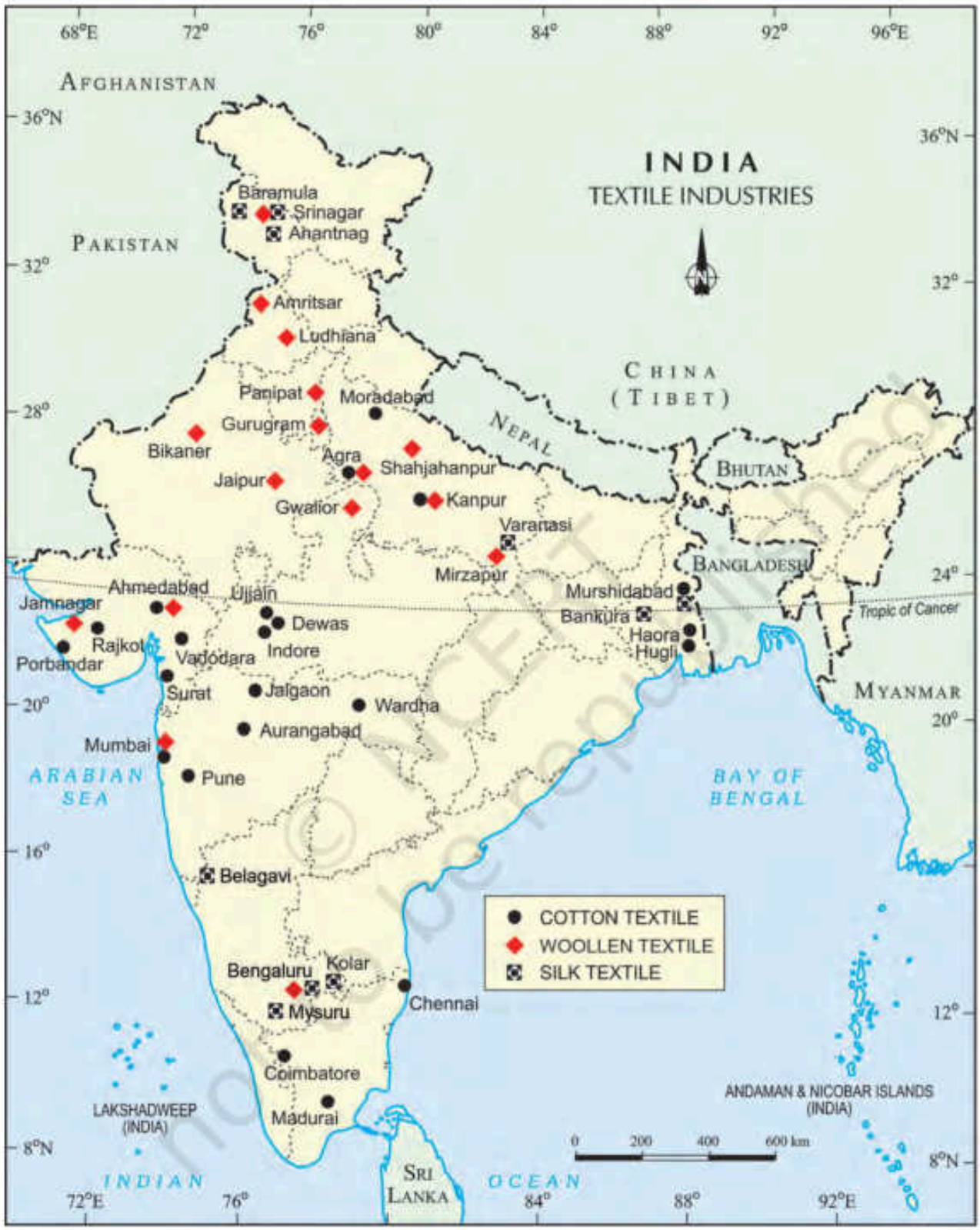
Jute Textiles

India is the largest producer of raw jute and jute goods and stands at second place as an exporter after Bangladesh. Most of the mills are located in West Bengal, mainly along the banks of the Hugli river, in a narrow belt.

The first jute mill was set up near Kolkata in 1855 at Rishra. After Partition in 1947, the jute mills remained in India but three-fourth of the jute producing area went to Bangladesh (erstwhile East Pakistan).



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India: Distribution of cotton, woollen and silk industries

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Factors responsible for their location in the Hugli basin are: proximity of the jute producing areas, inexpensive water transport, supported by a good network of railways, roadways and waterways to facilitate movement of raw material to the mills, abundant water for processing raw jute, cheap labour from West Bengal and adjoining states of Bihar, Odisha and Uttar Pradesh. Kolkata as a large urban centre provides banking, insurance and port facilities for export of jute goods.

Sugar Industry

India stands second as a world producer of sugar but occupies the first place in the production of *gur* and *khandsari*. The raw material used in this industry is bulky, and in haulage its sucrose content reduces. The mills are located in Uttar Pradesh, Bihar, Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Gujarat, Punjab, Haryana and Madhya Pradesh. Sixty per cent mills are in Uttar Pradesh and Bihar. This industry is seasonal in nature so, it is ideally suited to the cooperative sector. Can you explain why this is so?

In recent years, there is a tendency for the mills to shift and concentrate in the southern and western states, especially in Maharashtra. This is because the cane produced here has a higher sucrose content. The cooler climate also ensures a longer crushing season. Moreover, the cooperatives are more successful in these states.

Mineral-based Industries

Industries that use minerals and metals as raw materials are called mineral-based industries. Can you name some industries that would fall in this category?

Iron and Steel Industry

The iron and steel industry is the basic industry since all the other industries — heavy, medium and light, depend on it for their machinery. Steel is needed to manufacture a variety of engineering goods, construction material, defence, medical, telephonic, scientific equipment and a variety of consumer goods.

Activity

Make a list of all such goods made of steel that you can think of.

Production and consumption of steel is often regarded as the index of a country's development. Iron and steel is a heavy industry because all the raw materials as well as finished goods are heavy and bulky entailing heavy transportation costs. Iron ore, coking coal and lime stone are required in the ratio of approximately 4 : 2 : 1. Some quantities of manganese, are also required to harden the steel. Where should the steel plants be ideally located? Remember that the finished products also need an efficient transport network for their distribution to the markets and consumers.

Processes of Manufacture of Steel

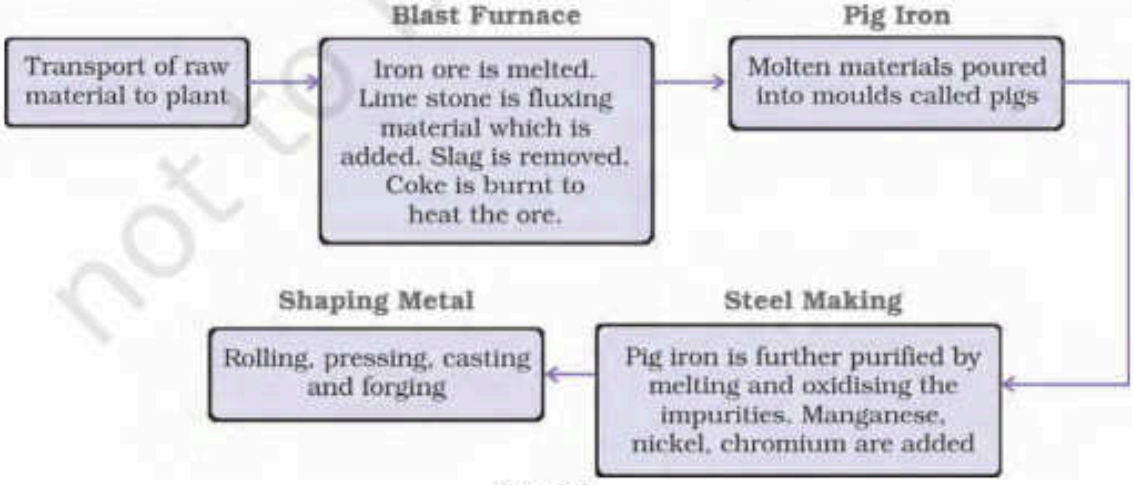


Fig. 6.2

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India: Iron and Steel Plants

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Chhotanagpur plateau region has the maximum concentration of iron and steel industries. It is largely, because of the relative advantages this region has for the development of this industry. These include, low cost of iron ore, high grade raw materials in proximity, cheap labour and vast growth potential in the home market.

Aluminium Smelting

Aluminium smelting is the second most important metallurgical industry in India. It is light, resistant to corrosion, a good conductor of heat, malleable and becomes strong when it is mixed with other metals. It is used to manufacture aircraft, utensils and wires. It has gained popularity as a substitute of steel, copper, zinc and lead in a number of industries.



Fig. 6.3: Strip coasting mill at smelter of NALCO

Aluminium smelting plants in the country are located in Odisha, West Bengal, Kerala, Uttar Pradesh, Chhattisgarh, Maharashtra and Tamil Nadu.

Bauxite, the raw material used in the smelters is a very bulky, dark reddish coloured rock. The flow chart given below shows the process of manufacturing aluminium. Regular supply of electricity and an assured source of raw material at minimum cost are the two prime factors for location of the industry.

Chemical Industries

The Chemical industry in India is fast growing and diversifying. It comprises both large and small scale manufacturing units. Rapid growth has been recorded in both inorganic and organic sectors. Inorganic chemicals include sulphuric acid (used to manufacture fertilizers, synthetic fibres, plastics, adhesives, paints, dyes stuffs), nitric acid, alkalies, soda ash (used to make glass, soaps and detergents, paper) and caustic soda. These industries are widely spread over the country.

Why do you think it is so?

Organic chemicals include petrochemicals, which are used for manufacturing of synthetic fibers, synthetic rubber, plastics, dye-stuffs, drugs and pharmaceuticals. Organic chemical

4 to 6 tonnes of bauxite ➡ 2 tonnes of alumina ➡ 1 tonne of aluminium

Fig. 6.4

Process of Manufacturing in Aluminium Industry

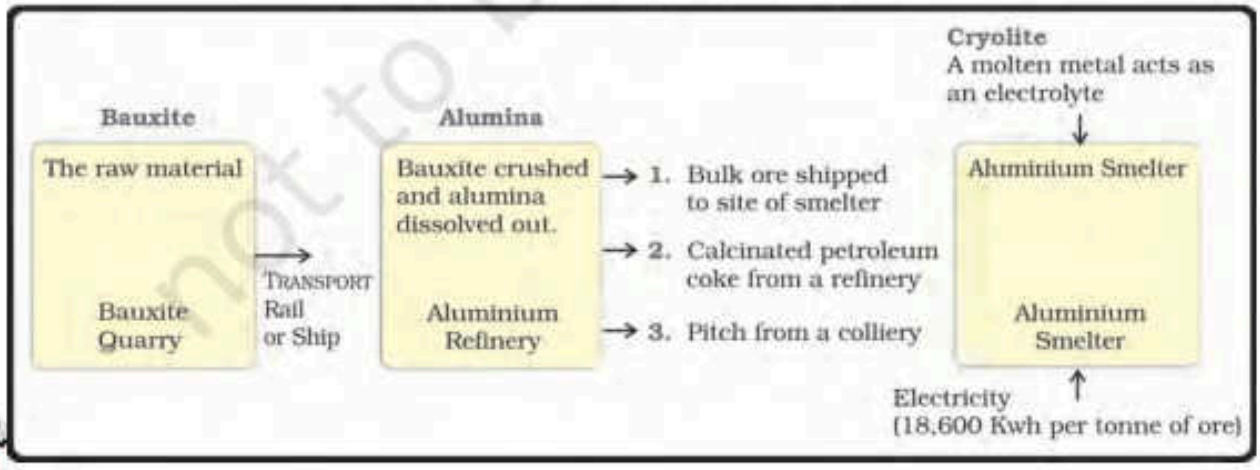


Fig. 6.5

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plants are located near oil refineries or petrochemical plants.

The chemical industry is its own largest consumer. Basic chemicals undergo processing to further produce other chemicals that are used for industrial application, agriculture or directly for consumer markets. Make a list of the products you are aware of.

Fertilizer Industry

The fertilizer industry is centred around the production of nitrogenous fertilizers (mainly urea), phosphatic fertilizers and ammonium phosphate (DAP) and complex fertilizers which have a combination of nitrogen (N), phosphate (P), and potash (K). The third, i.e. potash is entirely imported as the country does not have any reserves of commercially usable potash or potassium compounds in any form.

After the Green Revolution the industry expanded to several other parts of the country. Gujarat, Tamil Nadu, Uttar Pradesh, Punjab and Kerala contribute towards half of the fertilizer production. Other significant producers are Andhra Pradesh, Odisha, Rajasthan, Bihar, Maharashtra, Assam, West Bengal, Goa, Delhi, Madhya Pradesh and Karnataka.

Cement Industry

Cement is essential for construction activity such as building houses, factories, bridges, roads, airports, dams and for other commercial establishments. This industry requires bulky and heavy raw materials like limestone, silica and gypsum. Coal and electric power are needed apart from rail transportation.

Activity

Where would it be economically viable to set up the cement manufacturing units?

The industry has strategically located plants in Gujarat that have suitable access to the market in the Gulf countries.

Activity

Find out where the plants are located in other States of India. Find their names.

The first cement plant was set-up in Chennai in 1904. After Independence the industry expanded.

Automobile Industry

Automobiles provide vehicle for quick transport of good services and passengers. Trucks, buses, cars, motor cycles, scooters, three-wheelers and multi-utility vehicles are manufactured in India at various centres. After the liberalisation, the coming in of new and contemporary models stimulated the demand for vehicles in the market, which led to the healthy growth of the industry including passenger cars, two and three-wheelers. The industry is located around Delhi, Gurugram, Mumbai, Pune, Chennai, Kolkata, Lucknow, Indore, Hyderabad, Jamshedpur and Bengaluru.

Information Technology and Electronics Industry

The electronics industry covers a wide range of products from transistor sets to television, telephones, cellular telecom, telephone exchange, radars, computers and many other equipments required by the telecommunication industry. Bengaluru has emerged as the electronic capital of India. Other important centres for electronic goods are Mumbai, Delhi, Hyderabad, Pune, Chennai, Kolkata, Lucknow and Coimbatore. The major industry concentration is at Bengaluru, Noida, Mumbai, Chennai, Hyderabad and Pune. A major impact of this industry has been on employment generation. The continuing growth in the hardware and software is the key to the success of IT industry in India.



Fig. 6.6: Cable manufacturing facilities at HCL, Rupnarainpur (West Bengal)

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HOW TO COMPARE DIFFERENT COUNTRIES OR STATES?

You might ask if development can mean different things, how come some countries are generally called developed and others under-developed? Before we come to this, let us consider another question.

When we compare different things, they could have similarities as well as differences. Which aspects do we use to compare them? Let us look at students in the class itself. How do we compare different students? They differ in their height, health, talents and interests. The healthiest student may not be the most studious one. The most intelligent student may not be the friendliest one. So, how do we compare students? The criterion we may use depends on the purpose of comparison. We use different criterion to choose a sports team, a debate team, a music team or a team to organise a picnic. Still, if for some purpose, we have to choose the criterion for the all-round progress of children in the class, how shall we do it?

Usually we take one or more important characteristics of persons and compare them based on these characteristics. Of course, there can be differences about what are important characteristics that should form the basis of comparison: friendliness and spirit of cooperation, creativity or marks secured?

This is true of development too. **For comparing countries, their income is considered to be one of the most important attributes.** Countries with higher income are more developed than others with less income. This is based on the understanding that more income means more of all things that human beings need. Whatever people like, and should have, they will be able to get with greater income. So, greater income itself is considered to be one important goal.

Now, what is the income of a country? Intuitively, the income of the country is the income of all the residents of the country. This gives us the total income of the country.

However, for comparison between countries, total income is not such an useful measure. Since, countries have different populations, comparing total income will not tell us what an average person is likely to earn. Are people in one country better off than others in a different country? Hence, we compare the **average income** which is the total income of the country divided by its total population. The average income is also called **per capita income.**

In World Development Reports, brought out by the World Bank, this criterion is used in classifying countries. Countries with per capita income of US$ 66,500 per annum and above in 2024, are called high income or rich countries and those with per capita income of about US$ 2,300 or less are called low-income countries. India comes in the category of low middle income countries because its per capita income in 2024 was just about US$ 11,000 per annum. The rich countries, excluding countries of Middle East and certain other small countries, are generally called developed countries.

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Average Income

While 'averages' are useful for comparison, they also hide disparities

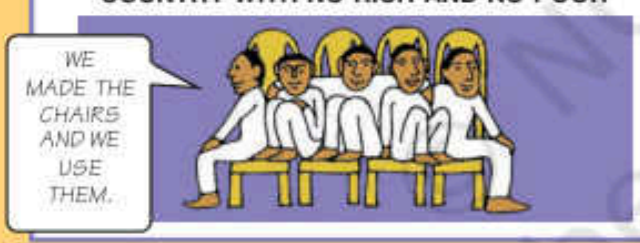
For example, let us consider two countries, A and B. For the sake of simplicity, we have assumed that they have only five citizens each. ***Based on data given in Table 1.2, calculate the average income for both the countries.***

TABLE 1.2 COMPARISON OF TWO COUNTRIES

Country	Monthly incomes of citizens (in Rupees)					
	I	II	III	IV	V	Average
Country A	9500	10500	9800	10000	10200	
Country B	500	500	500	500	48000	

Will you be equally happy to live in both these countries? Are both equally developed? Perhaps some of us may like to live in country B if we are assured of being its fifth citizen but if it is a lottery that decides our citizenship number then perhaps most of us will prefer to live in country A. Even though both the countries have identical average income, country A is preferred because it has more equitable distribution. In this country people are neither very rich nor extremely poor. On the other hand most citizens in country B are poor and one person is extremely rich. Hence, while average income is useful for comparison it does not tell us how this income is distributed among people.

COUNTRY WITH NO RICH AND NO POOR



COUNTRY WITH RICH AND POOR



LET'S WORK THESE OUT

1. Give three examples where an average is used for comparing situations.
2. Why do you think average income is an important criterion for development? Explain.
3. Besides size of per capita income, what other property of income is important in comparing two or more societies?
4. Suppose records show that the average income in a country has been increasing over a period of time. From this, can we conclude that all sections of the economy have become better? Illustrate your answer with an example.
5. From the text, find out the per capita income level of about 10-15 low-income countries as per World Development Reports.
6. Write a paragraph on your notion of what should India do, or achieve, to become a developed country.

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INCOME AND OTHER CRITERIA

When we looked at individual aspirations and goals, we found that people not only think of better income but also have goals such as security, respect for others, equal treatment, freedom, etc. in mind. Similarly, when we think of a nation or a region, we may, besides average income, think of other equally important attributes.

TABLE 1.3 PER CAPITA INCOME OF SELECT STATES

State	Per Capita Income for 2023–24 (in Rs)
Haryana	3,25,759
Kerala	2,81,001
Bihar	60,337

Source : Economic Survey 2024-25, Statistical Appendix, p. 32

What could these attributes be? Let us examine this through an example. Table 1.3 gives the per capita income of Haryana, Kerala and Bihar. Actually, these figures are of Per Capita Net State Domestic Product at Current Prices for 2023–24. Let us ignore what this complicated term exactly means. Roughly, we can take it to be the per capita income of the state. We find that of the three, Haryana has the highest per capita income and Bihar is at the bottom. This means that, on an average, a person in Haryana earned Rs 3,25,759 in one year whereas, on an average, a person in Bihar earned only around Rs 60,337. So, if per capita income were to be used as the measure of development, Haryana will be considered the most developed and Bihar the least developed state of the three. Now, let us look at certain other data pertaining to these states given in Table 1.4.

TABLE 1.4 SOME COMPARATIVE DATA ON HARYANA, KERALA AND BIHAR

State	Infant Mortality Rate per 1,000 live births (2020)	Literacy Rate % 2017–18	Net Attendance Ratio (per 100 persons) secondary stage (aged 15–17 years) 2017–18
Haryana	28	82	73
Kerala	6	94	94
Bihar	27	62	69

Sources : Economic Survey 2024–25, National Sample Survey Organisation (Report No. 585), National Statistical Office, Government of India; National Family Health Survey (NFHS-5) 2019-21 IIPS, Mumbai.

Explanation of some of the terms used in this table:

Infant Mortality Rate *(or* ***IMR****) indicates the number of children that die before the age of one year as a proportion of 1000 live children born in that particular year.*

Literacy Rate *measures the proportion of literate population in the 7-and-above age group.*

Net Attendance Ratio *is the total number of children of age group of 15-17 years attending school as a percentage of total number of children in the same age group.*

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What does this table show? The first column of the table shows that in Kerala, out of 1000 children born, 6 died before completing one year of age but in Haryana the proportion of children dying within one year of birth was 28, which is nearly three times more than that of Kerala. On the other hand, the per capita income of Haryana is more than that of Kerala as shown in Table 1.3. Just think of how dear you are to your parents, think of how every one is so happy when a child is born. Now, try to think of parents whose children die before they even celebrate their first birthday. How painful it must be to these parents? Next, note the year to which this data pertains. This was during 2018. So we are not talking of old times; it is 70 years after independence when our metro cities are full of high rise buildings and shopping malls!

The problem does not end with Infant Mortality Rate. The last column of Table 1.4 shows that about one-third of the children aged 15–17 years in Bihar are not attending school in secondary classes. This means that if you went to school in Bihar nearly one-third of your class friends would be missing. Those who could have been in school are not there! If this had happened to you, you would not be able to read what you are reading now.



Most babies require basic healthcare

PUBLIC FACILITIES

How is it that the average person in Haryana has more income than the average person in Kerala but lags behind in these crucial areas? The reason is — **money in your pocket cannot buy all the goods and services that you may need to live well.** So, income by itself is not a completely adequate indicator of material goods and services that citizens are able to use. For example, normally, your money cannot buy you a pollution-free environment or ensure that you get unadulterated medicines, unless you can afford to shift to a community that already has all these things. Money may also not be able to protect you from infectious diseases, unless the whole of your community takes preventive steps.



Actually for many of the important things in life the best way, also the cheapest way, is to provide these goods and services collectively. Just think – will it be cheaper to have collective security for the whole locality or for each house to have its own security staff? What if no one, other than you, in your village or locality is interested in studying? Would you be able to study? Not unless your parents could afford to send you to some private school elsewhere. So you are actually able to study because many other children also want to study and because many people believe that the government should open schools and provide other facilities so that all children have a chance to study. Even now, in many areas, children, particularly girls, are not able to go to high school because the government/society has not provided adequate facilities.

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Kerala has a low Infant Mortality Rate because it has adequate provision of basic health and educational facilities. Similarly, in some states, the Public Distribution System (PDS) functions well. Health and nutritional status of people of such states is certainly likely to be better.

LET'S WORK THESE OUT

1. Look at data in Tables 1.3 and 1.4. Is Haryana ahead of Kerala in literacy rate etc., as it is in terms of per capita income?
2. Think of other examples where collective provision of goods and services is cheaper than individual provision.
3. Does availability of good health and educational facilities depend only on amount of money spent by the government on these facilities? What other factors could be relevant?
4. In a study, it was found that in Tamil Nadu, 90 per cent of the people living in rural areas used a ration shop, whereas in West Bengal only 35 per cent of rural people did so. Where would people did better off and why?



ACTIVITY 2

Study Table 1.5 carefully and fill in the blanks in the following paragraphs. For this, you may need to make calculations based on the table.

TABLE 1.5 EDUCATIONAL ACHIEVEMENT OF RURAL POPULATION OF UTTAR PRADESH

Category	Male	Female
Literacy rate for rural population	76%	54%
Literacy rate for rural children in age group 10-14 years	90%	87%
Percentage of rural children aged 10-14 attending school	85%	82%

(a) The literacy rate for all age groups, including young and old, is _____ for rural males and _____ for rural females. However, it is not just that these many adults could not attend school but that there are _____ who are currently not in school.

(b) It is clear from the table that _____ % of rural girls and _____% of rural boys are not attending school. Therefore, illiteracy among children in the age group 10-14 is as high as _____% for rural females and _____% for rural males.

(c) This high level of illiteracy among _____ age group, even after more than 75 years of our independence, is most disturbing. In many other states also we are nowhere near realisation of the constitutional goal of free and compulsory education for all children up to the age of 14, which was expected to be achieved by 1960.

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ACTIVITY 3

One way to find out if we are properly nourished is to calculate what nutrition scientists call Body Mass Index (BMI). This is easy to calculate. Let each student in the class find out his or her weight and height. Take the weight of each student in kilograms (kg). Then, take the height by drawing up a scale on the wall and measuring accurately with the head straight. Convert the height recorded in centimeters into meters. Divide the weight in kg by the square of the height. The number you get is called BMI. Then, look at the BMI-for-Age tables given on pages 90–91. A student's BMI could be within the normal range or less than that (underweight) or more (obesity). For example, if a girl student is 14 years and 8 month old and the BMI is 15.2, then she is undernourished. Similarly, if the BMI of a boy aged 15 years and 6 months is 28, then he is overweight. Discuss the life situation, food and exercise habits of students, in general, without body shaming anyone.



HUMAN DEVELOPMENT REPORT

Once it is realised that even though the level of income is important, yet it is an inadequate measure of the level of development, we begin to think of other criterion. There could be a long list of such criterion but then it would not be so useful. What we need is a small number of the most important things. Health and education indicators, such as the ones we used in comparison of Kerala and Haryana, are among them. Over the past decade or so, health and education indicators have come to be widely used along with income as a measure of development. For instance, ***Human Development Report*** *published by UNDP compares countries based on the educational levels of the people, their health status and per capita income. It would be interesting to look at certain relevant data regarding India and its neighbours from Human Development Report 2025.*

TABLE 1.6 SOME DATA REGARDING INDIA AND ITS NEIGHBOURS FOR 2023

Country	Gross National Income (GNI) Per Capita (2021 PPP $)	Life Expectancy at birth	Mean Years of Schooling of People aged 25 and above	HDI Rank in the world (2021-22)
Sri Lanka	12,616	77.5	10.8	89
India	**9,047**	**72**	**6.9**	**130**
Myanmar	4,919	66.9	6.4	150
Pakistan	5,501	67.6	4.3	168
Nepal	4,726	70.4	4.5	145
Bangladesh	8,498	74.7	6.8	130

Source : Human Development Report, 2025, United Nations Development Programme, New York.

NOTES

1. *HDI stands for Human Development Index. HDI ranks in above table are out of 193 countries in all.*
2. *Life Expectancy at birth denotes, as the name suggests, average expected length of life of a person at the time of birth.*
3. *Per Capita Income is calculated in dollars for all countries so that it can be compared. It is also done in a way so that every dollar would buy the same amount of goods and services in any country.*

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Isn't it surprising that a small country in our neighbourhood, Sri Lanka, is much ahead of India in every respect and a big country like ours has such a low rank in the world? Table 1.6 also shows that though Nepal and Bangladesh have low per capita income than that of India, yet they are better than India in life expectancy.

Many improvements have been suggested in calculating HDI and many new components have been added to the Human Development Report but, by pre-fixing Human to Development, it has made it very clear that what is important in development is what is happening to citizens of a country. It is people, their health, their well being, that is most important.

Do you think there are certain other aspects that should be considered in measuring human development?

SUSTAINABILITY OF DEVELOPMENT

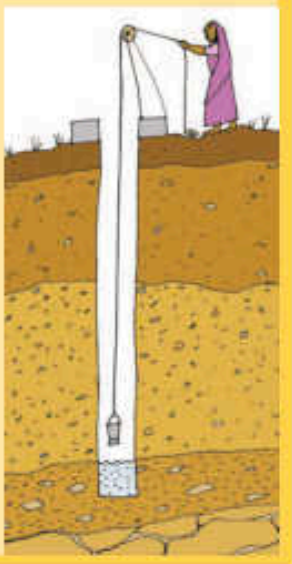
Suppose for the present that a particular country is quite developed. We would certainly like this level of development to go up further or at least be maintained for future generations. This is obviously desirable. However, since the second half of the twentieth century, a number of scientists have been warning that the present type, and levels, of development are not sustainable.

> *"We have not inherited the world from our forefathers — we have borrowed it from our children."*

LET'S UNDERSTAND WHY THIS IS SO THROUGH THE FOLLOWING EXAMPLE:

Example 1: Groundwater in India

"Recent evidence suggests that the groundwater is under serious threat of overuse in many parts of the country. About 300 districts have reported a water level decline of over 4 metres during the past 20 years. Nearly one-third of the country is overusing their groundwater reserves. In another 25 years, 60 per cent of the country would be doing the same if the present way of using this resource continues. Groundwater overuse is particularly found in the agriculturally prosperous regions of Punjab and Western U.P., hard rock plateau areas of central and south India, some coastal areas and the rapidly growing urban settlements."



(a) Why groundwater is overused?

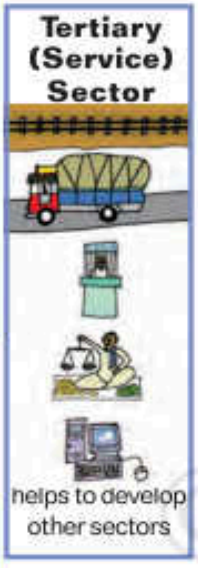
(b) Can there be development without overuse?

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We begin by looking at different kind of economic activities.

There are many activities that are undertaken by directly using natural resources. Take, for example, the cultivation of cotton. It takes place within a crop season. For the growth of the cotton plant, we depend **mainly, but not entirely,** on natural factors like rainfall, sunshine and climate. The product of this activity, cotton, is a natural product. Similarly, in the case of an activity like dairy, we are dependent on the biological process of the animals and availability of fodder etc. The product here, milk, also is a natural product. Similarly, minerals and ores are also natural products. When we produce a good by exploiting natural resources, it is an activity of the **primary sector**. Why primary? This is because it forms the base for all other products that we subsequently make. Since most of the natural products we get are from agriculture, dairy, fishing, forestry, this sector is also called **agriculture and related sector**.







The **secondary sector** covers activities in which natural products are changed into other forms through ways of manufacturing that we associate with industrial activity. It is the next step after primary. The product is not produced by nature but has to be made and therefore some process of manufacturing is essential. This could be in a factory, a workshop or at home. For example, using cotton fibre from the plant, we spin yarn and weave cloth. Using sugarcane as a raw material, we make sugar or *gur*. We convert earth into bricks and use bricks to make houses and buildings. Since this sector gradually became associated with the different kinds of industries that came up, it is also called as **industrial** sector.

After primary and secondary, there is a third category of activities that falls under **tertiary sector** and is different from the above two. These are activities that help in the development of the primary and secondary sectors. These activities, by themselves, do not produce a good but they are an aid or a support for the production process. For example, goods that are produced in the primary or secondary sector would need to be transported by trucks or trains and then sold in wholesale and retail shops. At times, it may be necessary to store these in godowns. We also may need to talk to others over telephone or send letters (communication) or borrow money from banks (banking) to help production and trade. Transport, storage, communication, banking, trade are some examples of tertiary activities. Since these activities generate services rather than goods, the tertiary sector is also called the **service sector**.

Service sector also includes some essential services that may not directly help in the production of goods. For example, we require teachers, doctors, and those who provide personal services such as washermen, barbers, cobblers, lawyers, and people to do administrative and accounting works. In recent times, certain new services based on information technology such as internet cafe, ATM booths, call centres, software companies etc have become important.

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Economic activities, though, are grouped into three different categories, are highly **interdependent.** Let us look at some examples.

TABLE 2.1 EXAMPLES OF ECONOMIC ACTIVITIES

EXAMPLE	WHAT DOES THIS SHOW?
Imagine what would happen if farmers refuse to sell sugarcane to a particular sugar mill. The mill will have to shut down.	This is an example of the secondary or industrial sector being dependent on the primary.
Imagine what would happen to cotton cultivation if companies decide not to buy from the Indian market and import all cotton they need from other countries. Indian cotton cultivation will become less profitable and the farmers may even go bankrupt, if they cannot quickly switch to other crops. Cotton prices will fall.	
Farmers buy many goods such as tractors, pumpsets, electricity, pesticides and fertilisers. Imagine what would happen if the price of fertilisers or pumpsets go up. Cost of cultivation of the farmers will rise and their profits will be reduced.	
People working in industrial and service sectors need food. Imagine what would happen if there is a strike by transporters and lorries refuse to take vegetables, milk, etc. from rural areas. Food will become scarce in urban areas whereas farmers will be unable to sell their products.	

LET'S WORK THESE OUT

1. Complete the above table to show how sectors are dependent on each other.
2. Explain the difference between primary, secondary and tertiary sectors using examples other than those mentioned in the text.
3. Classify the following list of occupations under primary, secondary and tertiary sectors:
   - Tailor
   - Basket weaver
   - Flower cultivator
   - Milk vendor
   - Fishermen
   - Priest
   - Courier
   - Workers in match factory
   - Moneylender
   - Gardener
   - Potter
   - Bee-keeper
   - Astronaut
   - Call centre employee
4. Students in a school are often classified into primary and secondary or junior and senior. What is the criterion that is used? Do you think this is a useful classification? Discuss.

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COMPARING THE THREE SECTORS

The various production activities in the primary, secondary and tertiary sectors produce a very large number of goods and services. Also, the three sectors have a large number of people working in them to produce these goods and services. The next step, therefore, is to see how much goods and services are produced and how many people work in each sector. In an economy there could be one or more sectors which are dominant in terms of total production and employment, while other sectors are relatively small in size.

How do we count the various goods and services and know the total production in each sector?

With so many thousands of goods and services produced, you might think this is an impossible task! Not only would the task be enormous, you might also wonder how we can add up cars and computers and nails and furniture. It won't make sense!!!

You are right in thinking so. To get around this problem, economists suggest that the values of goods and services should be used rather than adding up the actual numbers. For example, if 10,000 kgs of wheat is sold at Rs 20 per kg, the value of wheat will be Rs 2,00,000. The value of 5000 coconuts at Rs 15 per coconut will be Rs 75,000. Similarly, the value of goods and services in the three sectors are calculated, and then added up.



Remember, there is one precaution one has to take. Not every good (or service) that is produced and sold needs to be counted. It makes sense only to include the **final goods and services**. Take, for instance, a farmer who sells wheat to a flour mill for Rs 20 per kg. The mill grinds the wheat and sells the flour to a biscuit company for Rs 25 per kg. The biscuit company uses the flour and things such as sugar and oil to make four packets of biscuits. It sells biscuits in the market to the consumers for Rs 80 (Rs 20 per packet). Biscuits are the final goods, i.e., goods that reach the consumers.

Why are only 'final goods and services' counted? In contrast to final goods, goods such as wheat and the wheat flour in this example are intermediate goods. Intermediate goods are used up in producing final goods and services. The value of final goods **already includes** the value of all the intermediate goods that are used in making the final good. Hence, the value of Rs 80 for the biscuits (final good) already includes the value of flour (Rs 25). Similarly, the value of all other intermediate goods would have been included. To count the value of the flour and wheat separately is therefore not correct because then we would be counting the value of the same things a number of times. First as wheat, then as flour and finally as biscuits.

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The value of final goods and services produced in each sector during a particular year provides the total production of the sector for that year. And the sum of production in the different sectors gives what is called the **Gross Domestic Product (GDP)** of a country. It is the value of all final goods and services produced **within a country** during a particular year. GDP shows how big the economy is.

In India, the mammoth task of measuring GDP is undertaken by a central government ministry. This Ministry, with the help of various government departments of all the Indian states and union territories, collects information relating to total volume of goods and services and their prices and then estimates the GDP.

Recently Indian Government began to bring out the contribution of three sectors towards Gross Value Added (GVA) in place of the contribution of three sectors towards Gross Domestic Product to be at par with global practices. The GVA measures the contribution of three sectors of an economy after adjusting for taxes and subsidies. You will learn more about the GVA and the difference between GDP and GVA in higher classes. Here we will only understand how goods and services produced in these three sectors—primary, secondary and tertiary sectors have changed.

Historical Change in Sectors

Generally, it has been noted from the histories of many, now developed, countries that at initial stages of development, primary sector was the most important sector of economic activity.

As the methods of farming changed and agriculture sector began to prosper, it produced much more food than before. Many people could now take up other activities. There were increasing number of craft-persons and traders. Buying and selling activities increased many times. Besides, there were also transporters, administrators, army etc. However, at this stage, most of the goods produced were natural products from the primary sector and most people were also employed in this sector.

Over a long time (more than hundred years), and especially because new methods of manufacturing were introduced, factories came up and started expanding. Those people who had earlier worked on farms now began to work in factories in large numbers. They were forced to do so as you read in history chapters. People began to use many more goods that were produced in factories at cheap rates. Secondary sector gradually became the most important in total production and employment. Hence, over time, a shift had taken place. This means that the importance of the sectors had changed.

In the past 100 years, there has been a further shift from secondary to tertiary sector in developed countries. The service sector has become the most important in terms of total production. Most of the working people are also employed in the service sector. This is the general pattern observed in developed countries.

What is the total production and employment in the three sectors in India? Over the years have there been changes similar to the pattern observed for the developed countries? We shall see in the next section.

LET'S WORK THESE OUT

1. What does the history of developed countries indicate about the shifts that have taken place between sectors?
2. Correct and arrange the important aspects for calculating GDP from this Jumble.

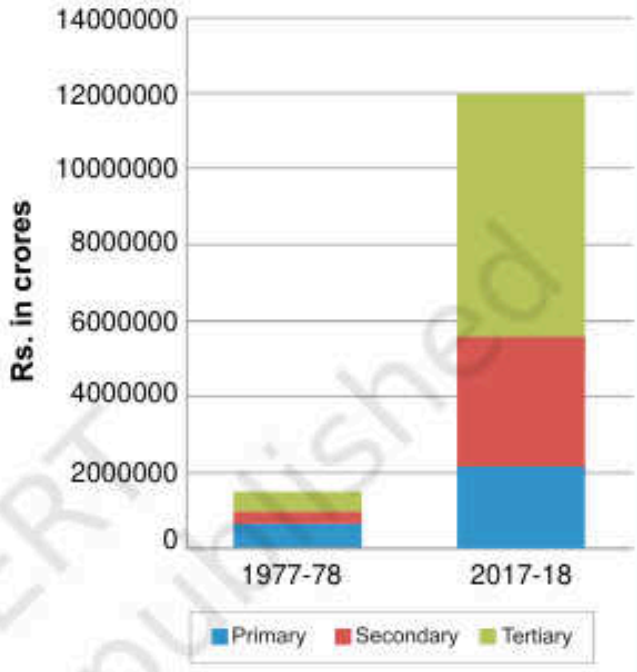
   To count goods and services we add the numbers that are produced. We count all those that were produced in the last five years. Since we shouldn't leave out anything we add up all these goods and services.
3. Discuss with your teacher how you could calculate the total value of a good or service by using the method of value added at each stage.

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PRIMARY, SECONDARY AND TERTIARY SECTORS IN INDIA

Graph 1 shows the production of goods and services in the three sectors. This is shown for two years, 1977–78 and 2017–18. We have used the data for these two years because the data are comparable and authentic. You can see how the total production has grown over the forty years.

Graph 1: GVA by Primary, Secondary and Tertiary Sectors



LET'S WORK THESE OUT

Answer the following questions by looking at the graph:

1. Which was the largest producing sector in 1977–78?
2. Which is the largest producing sector in 2017–18?
3. Can you say which sector has grown the most over forty years?
4. What was the GDP of India in 2017–18?

What does the comparison between 1977–78 and 2017–18 show? What conclusions can we draw from the comparison? Let's find out.

Rising Importance of the Tertiary Sector in Production

Over the forty years between 1977–78 and 2017–18, while production in all the three sectors has increased, it has increased the most in the tertiary sector. As a result, in the year 2017–18, the tertiary sector has emerged as the largest producing sector in India replacing the primary sector.

Why is the tertiary sector becoming so important in India? There could be several reasons.

First, in any country several services such as hospitals, educational institutions, post and telegraph services, police stations, courts, village administrative offices, municipal corporations, defence, transport, banks, insurance companies, etc. are required. These can be considered as *basic services*. In a developing country the government has to take responsibility for the provision of these services.

Second, the development of agriculture and industry leads to the development of services such as

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transport, trade, storage and the like, as we have already seen. Greater the development of the primary and secondary sectors, more would be the demand for such services.

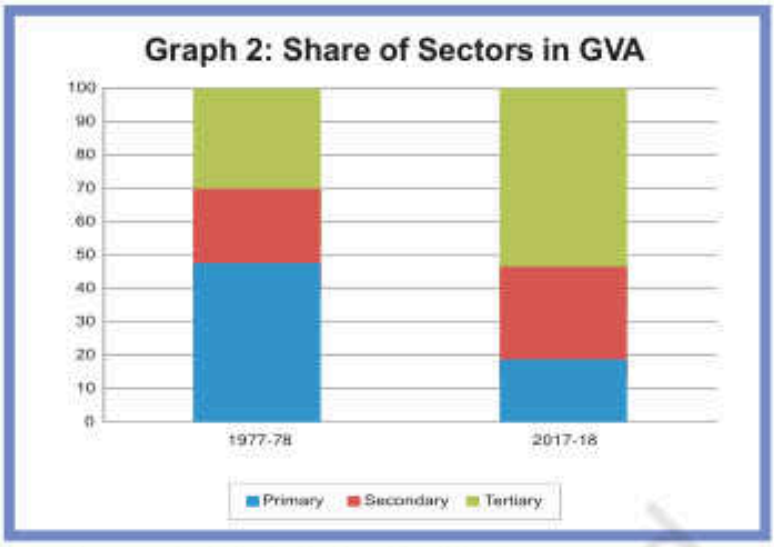
Third, as income levels rise, certain sections of people start demanding many more services like eating out, tourism, shopping, private hospitals, private schools, professional training etc. You can see this change quite sharply in cities, especially in big cities.

Fourth, over the past decade or so, certain new services such as those based on information and communication technology have become important and essential. The production of these services has been rising rapidly. In Chapter 4, we shall see examples of these new services and the reasons for their expansion.

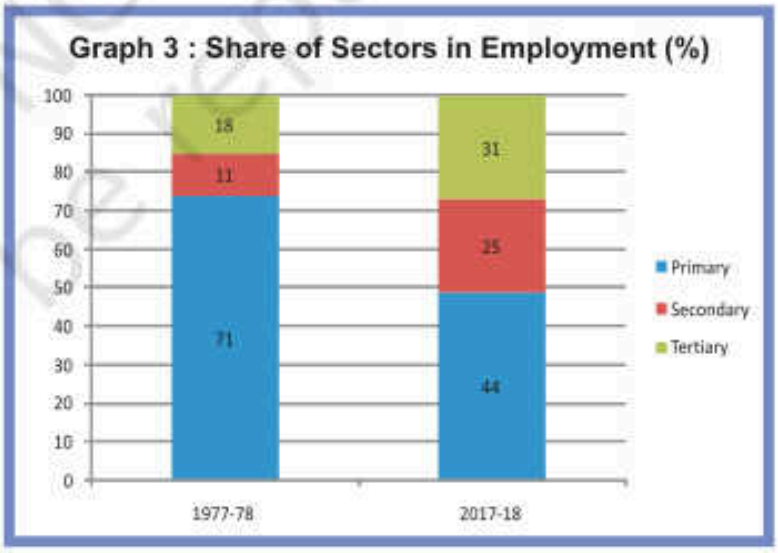
However, you must remember that not all of the service sector is growing equally well. Service sector in India employs many different kinds of people. At one end there are a limited number of services that employ highly skilled and educated workers. At the other end, there are a very large number of workers engaged in services such as small shopkeepers, repair persons, transport persons, etc. These people barely manage to earn a living and yet they perform these services because no alternative opportunities for work are available to them. Hence, only a part of this sector is growing in importance. You shall read more about this in the next section.

Where are most of the people employed?

Graph 2 presents percentage share of the three sectors in GVA. Now you can directly see the changing importance of the sectors over the forty years.



A remarkable fact about India is that while there has been a change in the share of the three sectors in GVA, a similar shift has not taken place in employment. Graph 3 shows the share of employment in the three sectors in 1977-78 and 2017-18. The primary sector continues to be the largest employer even now.



Why didn't a similar shift out of primary sector happen in case of employment? It is because not enough jobs were created in the secondary and tertiary sectors. Even

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though industrial output or the production of goods went up by more than nine times during the period, employment in the industry went up by around three times. The same applies to the tertiary sector as well. While production in the service sector rose by 14 times, employment in the service sector rose around five times.

As a result, more than half of the workers in the country are working in the primary sector, mainly in agriculture, producing only about one sixth of the GVA. In contrast to this, the secondary and tertiary sectors produce the rest of the produce whereas they employ less about half the people. Does this mean that the workers in agriculture are not producing as much as they could?

What it means is that there are more people in agriculture than is necessary. So, even if you move a few people out, production will not be affected. In other words, workers in the agricultural sector are **underemployed.**



For instance, take the case of a small farmer, Laxmi, owning about two hectares of unirrigated land dependent only on rain and growing crops, like *jowar* and *arhar*. All five members of her family work in the plot throughout the year. Why? They have nowhere else to go for work. You will see that everyone is working, none remains idle, but in actual fact, their **labour effort** gets divided. Each one is doing some work but no one is fully employed. This is the situation of underemployment, where people are apparently working but all of them are made to work less than their potential. **This kind of underemployment is hidden in contrast to someone who does not have a job and is clearly visible as unemployed. Hence, it is also called disguised unemployment.**

Now, supposing a landlord, Sukhram, comes and hires one or two members of the family to work on his land. Laxmi's family is now able to earn some extra income through wages. Since you do not need five people to look after that small plot, two people moving out does not affect production on their farm. In the above example, two people may move to work in a factory. Once again the earnings of the family would increase and they would also continue to produce as much from their land.

There are lakhs of farmers like Laxmi in India. This means that even if we remove a lot of people from agricultural sector and provide them with proper work elsewhere, agricultural production will not suffer. The incomes of the people who take up other work would increase the total family income.

This underemployment can also happen in other sectors. For example there are thousands of casual workers in the service sector in urban areas who search for daily employment. They are employed as painters, plumbers, repair persons and others doing odd jobs. Many of them don't find work everyday. Similarly, we see other people of the service sector on the street pushing a cart or selling something where **they may spend the whole day but earn very little.** They are doing this work because they do not have better opportunities.

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